



Article

AI-Driven Optimization of Sanitary Facilities in Office Buildings: A Machine Learning Approach Using LSTM Neural Networks

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Abstract: This study introduces an artificial intelligence-based approach to planning sanitary facilities in buildings, with a focus on long short-term memory neural networks. By examining factors like occupancy patterns, daily fluctuations, and building-specific characteristics, the model delivers data-driven guidance for fixture allocation. Validation was carried out using empirical data from a Singaporean office building, resulting in strong predictive performance and improved operational efficiency. The long short-term memory model surpassed traditional methods, with a 39.26% boost in mean absolute percentage error over queueing theory. In a real-world scenario, its predictions were within 2.63–4.76% of actual needs, whereas prescriptive codes deviated by 12.07–19.05%. Sensitivity analysis showed that total occupancy, time of day, and gender ratio exerted the greatest influence, enabling gender-specific recommendations. The findings also indicate a potential 15% cut in overall fixtures with no compromise in service quality, providing considerable cost and space savings. By uniting advanced computational modeling with real data, this research demonstrates how artificial intelligence can elevate sanitary facility planning toward evidence-based decision-making, efficient resource use, and higher user satisfaction. The results may influence policy, regulatory standards, and performance-oriented design in contemporary buildings.

Keywords: sanitary facility planning; machine learning; LSTM neural networks; predictive modeling; facility optimization; building management



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1. Introduction

Sanitary facilities are more than an afterthought in building design; they significantly affect occupant comfort, health, and resource use [1]. Historically, building codes—like New Zealand’s Building Code (Clause G1 Personal Hygiene) [2], the United Kingdom’s Building Regulations—Approved Document G [3], and the International Plumbing Code in the United States [4]—have guided fixture requirements. However, these guidelines often rely on static assumptions that fail to capture flexible work arrangements or varying daily usage. Thus, many facilities end up overbuilt or underprepared for peak demands.

Traditional approaches based on queueing theory [5,6] further illustrate the complexity. While mathematically sound, they tend to overlook the wide range of human behaviors involved in restrooms. This highlights the need for new approaches that adapt to modern building use, evolving demographics, and shifting occupancy patterns.

While existing approaches using queueing theory and prescriptive codes provide basic guidelines, they fail to account for dynamic usage patterns, cultural factors, and evolving

workplace demographics. There is a critical need for a data-driven, adaptive approach that can capture these complex interactions and provide more accurate predictions for sanitary facility requirements.

Current approaches to sanitary facility planning reveal significant gaps between theoretical models and real-world requirements. While existing methods based on queueing theory and prescriptive codes provide foundational guidelines, they exhibit several critical limitations. Traditional queueing models assume steady-state conditions and uniform arrival patterns, failing to capture the dynamic nature of modern workplace occupancy. Prescriptive codes, typically based on historical data and simplified occupancy assumptions, cannot adapt to emerging workplace trends such as flexible hours, remote work policies, and varying cultural practices.

Furthermore, existing approaches demonstrate three key shortcomings:

First, current methodologies lack the ability to account for temporal variations in facility usage. Peak demand periods, seasonal fluctuations, and event-driven usage spikes remain poorly addressed by static models. For instance, traditional calculations based on total occupancy fail to consider how usage patterns vary throughout the day or how special events might impact demand.

Second, contemporary workplace demographics and cultural factors receive insufficient consideration in existing models. The increasing diversity of office populations, varying cultural norms regarding facility usage, and gender-specific requirements are not adequately captured by current approaches. This oversight leads to potential mismatches between provided facilities and actual user needs.

Third, the emergence of flexible working arrangements and hybrid office models has introduced new complexities in occupancy patterns that traditional methods were not designed to handle. The rigid assumptions underlying current models become increasingly inappropriate as workplaces evolve toward more dynamic and adaptable environments.

There is, therefore, a critical need for a data-driven, adaptive approach that can perform the following:

- Capture and respond to dynamic usage patterns in real-time;
- Account for cultural factors and demographic variations;
- Adapt to evolving workplace trends and occupancy patterns;
- Provide more accurate predictions for fixture requirements based on actual usage data;
- Optimize resource allocation while maintaining user satisfaction.

This research gap presents an opportunity to develop a more sophisticated, AI-driven approach that can integrate multiple data sources, learn from usage patterns, and provide more accurate and adaptable recommendations for sanitary facility planning. The proposed LSTM-based framework addresses these limitations by offering a dynamic, data-driven solution that can evolve with changing workplace needs while maintaining high standards of service quality.

In Singapore, a regulatory void for office-specific sanitary facility standards has led to reliance on either broad international codes or personal heuristics. Under the National Environment Agency's (NEA) Code of Practice for Environmental Health 2024 [7], no direct guidance exists for offices, prompting designers to borrow from external sources that may not reflect Singapore's local norms. The Singtel Serangoon TEPL office building project in Singapore addressed this gap by examining cultural practices, workplace habits, and actual occupancy trends to form more precise, context-driven guidelines.

This research combined international codes, queueing theory, and region-specific insights. By probing cultural norms and dynamic occupancy, it became clear that existing rules and models sometimes fall short when applied to a local setting. These observations

drove the need for more sophisticated methods, specifically LSTM neural networks, to generate better-informed recommendations.

Recent years have seen AI revolutionize problem-solving in facility management. Unlike rigid models, machine learning (ML) algorithms embrace nonlinear interactions and can adjust to new data streams. LSTM neural networks [8,9], designed for sequential inputs, show particular promise when forecasting restroom usage patterns, given their ability to handle time-dependent data and unpredictable behaviors. Building on these advances, this research develops and tests an AI-based framework to optimize sanitary fixture requirements in office buildings in Singapore. Drawing on data from the Singtel Serangoon TEPL building at 1 Serangoon North Avenue 5, Singapore, it aims to build a predictive model that accounts for time-based, demographic, and environmental factors; compare AI-based forecasts against traditional methods; pinpoint critical factors shaping restroom usage; and illustrate practical benefits of the model in a working environment.

This research makes several significant contributions to the field of sanitary facility planning. We present a novel LSTM-based neural network architecture specifically designed for sanitary facility optimization. This framework integrates temporal dependencies, occupancy patterns, and environmental factors to provide more accurate predictions than traditional methods. Our approach achieves a 39.26% improvement in mean absolute percentage error compared to conventional queueing theory models, demonstrating significant advancement in prediction accuracy for facility planning.

We validate our approach using comprehensive empirical data from the Singtel Serangoon TEPL office building, supplemented by testing on two additional office complexes. This validation demonstrates that our model can achieve prediction accuracies within 2.63–4.76% of actual usage patterns, compared to 12.07–19.05% deviations observed with prescriptive codes. The framework's effectiveness is verified across different building sizes and occupancy profiles, confirming its practical applicability in real-world scenarios.

Through systematic sensitivity analysis, we identify and quantify the key factors influencing sanitary facility demand. Our research reveals that total building occupancy (sensitivity coefficient: 0.842), time of day (0.731), and gender ratio (0.695) are the most significant determinants of fixture requirements. These insights enable more precise resource allocation by allowing facility planners to prioritize the most impactful factors. They facilitate gender-specific optimization, address critical gaps in current planning approaches, and help predict and manage peak demand periods more effectively. This analysis provides a scientific basis for dynamic facility management strategies, potentially reducing wait times by up to 40%, while supporting cost-effective design decisions by identifying where investments will have the greatest impact.

Our research demonstrates potential resource savings of up to 15% in fixture requirements while maintaining service quality, with a maximum 60-s wait time for 90% of users. This optimization translates to an approximately SGD 280,000 reduction in construction costs, annual maintenance savings of SGD 42,000, and recovery of approximately 45 m² of valuable floor space. These findings provide a quantitative foundation for more efficient facility planning and resource allocation in modern office buildings.

The significance of these contributions extends beyond theoretical advancement, offering practical tools and insights for architects, facility managers, and building planners. Our framework represents a shift from static, prescriptive approaches to dynamic, data-driven decision-making in sanitary facility planning. The methodology developed here could potentially influence building codes and standards, leading to more efficient resource allocation and improved user satisfaction. Furthermore, the insights gained about usage patterns and influential factors contribute to our understanding of how modern workplaces function and evolve.

The remainder of this paper is organized as follows: Section 2 reviews related work in sanitary facility planning and machine learning applications. Section 3 describes our methodology, including data collection, feature engineering, and model development. Section 4 presents the results and analysis of our approach. Section 5 discusses implications and limitations of the findings, and Section 6 concludes with recommendations for future research directions.

2. Literature Review

2.1. Prescriptive Codes—Comparative Analysis of International Toilet Standards

Prescriptive codes have long provided the backbone for establishing fixture counts across multiple regions. Frameworks like the International Plumbing Code [4], the UK's Approved Document G [3], and Australia's National Construction Code [10] typically define basic compliance requirements. While straightforward, these standards can differ substantially, reflecting each location's cultural norms and regulatory outlook.

However, international standards diverge significantly due to cultural practices, regulatory priorities, and design philosophies. Greed [1] shed light on disparities in Western regulations, especially regarding gender equity and accessibility. The UK's Approved Document G [3] strives to address these issues, while Australia's NCC [10] offers a more performance-based angle, contrasting with the IPC's [4] focus on rigid guidelines. Despite their utility, prescriptive rules are as follows:

- Rely on static models that may miss flexible work hours or variable occupancy [11];
- Struggle with unique cultural practices or peak events [1];
- Risk over- or under-providing, leading to cost burdens or user frustration [12];
- Often refer to outdated assumptions, overlooking workplace shifts like remote work or evolving gender norms [13].

Studies such as those by Ching and Winkel [11] reinforce the call for flexible, data-powered frameworks in response to global variations and changing usage patterns.

2.2. Queueing Theory Models for Toilet Facilities

Queueing theory has a long history in analyzing restroom capacity. Early M/M/c formulations by Haight [14] shaped the basic concept of arrival rates and service times, while Davidson and Courtney [13] introduced M/G/c approaches to capture more realistic patterns. Subsequent improvements integrated are as follows:

- Gender-based factors (Kira [15]);
- Time-sensitive variables (Rothausen-Vange et al. [12]).

However, these models tend to oversimplify arrival rates, ignore cultural or behavioral nuances, and need constant parameter fine-tuning [11,13,16]. Such constraints highlight the necessity for more adaptive strategies—ones that machine learning can support.

2.3. Applications of Machine Learning in Facilities Management

2.3.1. Evolution of Neural Network Architectures

Long short-term memory (LSTM) networks represent a specialized architecture within the family of recurrent neural networks (RNNs), specifically designed to address the vanishing gradient problem that hampers traditional RNNs in capturing long-term dependencies. The LSTM architecture, first introduced by Hochreiter and Schmidhuber [17], incorporates a sophisticated gating mechanism that allows the network to selectively retain or forget information over extended sequences.

Recent advances have expanded upon this foundation. Naveed et al. [18] demonstrated how transformer-infused recurrent networks achieve superior accuracy in predict-

ing environmental patterns, with their TIR model attaining an R^2 value of 0.9983. Their integration of BiLSTM for encoding and GRU for decoding, enhanced by attention mechanisms, provides an architectural pattern particularly suitable for complex temporal data in infrastructure management.

2.3.2. Addressing Limited Training Data Constraints

The challenge of limited training data emerges as a significant consideration in facility management applications. Iqbal et al. [19] examined this constraint through systematic comparison of learning algorithms for small datasets, revealing that adaptive gradient methods achieve better accuracy and reduced training time compared to conventional approaches. Their findings on hyperparameter optimization and model architecture selection directly inform methodological choices in facility prediction tasks.

2.3.3. Successful Applications in Building Management

Recent implementations demonstrate the versatility of advanced neural networks across various building management domains. Liu et al. [20] implemented an LSTM-based framework for building energy consumption prediction, achieving a 23% improvement in accuracy over traditional time-series models. Their system successfully captured complex patterns in HVAC usage across commercial buildings.

In solar energy applications, Zhang et al. [21] employed a modified LSTM architecture incorporating metaheuristic optimization techniques, demonstrating a 15% improvement in prediction accuracy over conventional networks. Wang and Chen [22] achieved 92% accuracy in structural response prediction for train-bridge systems, while Martinez et al. [23] reported a MAPE of 5.8% for occupancy prediction in commercial buildings.

2.3.4. Applications in Sanitary Facility Management

The success of LSTM networks in these domains suggests promising applications for sanitary facility optimization. Several key parallels exist between these applications and sanitary facility management. The temporal pattern recognition capabilities proven in energy management translate directly to restroom usage, which exhibits similar short-term and long-term patterns. The multi-variable dependencies handled in photovoltaic production forecasting mirror the complex interrelationships between occupancy, time of day, and gender distribution in sanitary facility usage. Furthermore, the real-time adaptation capabilities demonstrated in seismic response prediction prove valuable for adjusting to varying occupancy patterns and special events in facility management.

These successful applications provide a strong foundation for extending LSTM technology to sanitary facility planning. The demonstrated capabilities in handling complex temporal patterns, multiple variables, and real-time adaptation align perfectly with the challenges of optimizing restroom facilities in modern office buildings.

These successful applications provide a strong foundation for extending LSTM technology to sanitary facility planning. The demonstrated capabilities in handling complex temporal patterns, multiple variables, and real-time adaptation align perfectly with the challenges of optimizing restroom facilities in modern office buildings. Such methods excel at handling diverse inputs, scaling well, and uncovering usage patterns that simpler models miss. Nonetheless, broader data gathering and proper validation remain crucial to confirm their reliability in restroom planning.

2.4. Synthesis of Current Approaches

While both queueing theory and machine learning approaches have advanced our understanding of facility planning, significant opportunities remain for integration and improvement. Previous research [6,16,24,25] has demonstrated that theoretical models

often diverge from real-world user experiences, particularly in dynamic, time-sensitive settings. This research addresses these limitations by merging the strengths of queueing theory, prescriptive guidelines, and cutting-edge machine learning to propose a more adaptive and context-aware pathway for sanitary facility planning. Our methodology, detailed in the following section, builds upon these foundations while addressing the identified limitations in current approaches.

3. Methodology

3.1. Overview of Proposed Machine Learning Approach

This study adopts a long short-term memory (LSTM) neural network, a special class of recurrent neural network (RNN) designed to handle time-series data by capturing both near-term and long-term dependencies. Such adaptability is crucial for modeling dynamic restroom usage, where facility demand can shift throughout the day.

The methodological workflow encompasses five primary stages:

1. Data Collection and Preprocessing—Compiling both empirical and synthetic datasets to reflect a range of usage scenarios.
2. Feature Engineering—Identifying and extracting key variables, including occupancy levels, gender distribution, and temporal factors.
3. Model Development and Training—Designing and refining the LSTM architecture using the preprocessed dataset.
4. Hyperparameter Tuning—Systematically optimizing the model configuration to elevate predictive accuracy.
5. Performance Evaluation—Measuring the final model's performance against standard methods like queueing theory.

The model's input features capture occupancy, time of day, and gender proportions, while its output focuses on estimating the required counts of toilets, urinals, and washbasins for each gender.

3.2. Dataset Description and Analysis

3.2.1. Data Collection Methodology

The dataset was collected from the Singtel Serangoon TEPL office building over a 12-month period (January 2024 to December 2024). The data collection covered five floors (levels 2–6), encompassing 2606 total occupants and 223 sanitary fixtures, with continuous 24/7 temporal coverage at 1-min sampling intervals. The dataset is provided as Supplementary Materials.

Data collection utilized multiple monitoring systems, including automated occupancy sensors at entry/exit points, IoT-enabled fixture usage monitors, and building management system logs. Monthly manual validation surveys supplemented the automated data collection to ensure accuracy and reliability.

3.2.2. Dataset Composition

The primary dataset comprises three key components. First, the occupancy data covers a total building population of 2606 occupants, with a gender distribution of 54% male (1403) and 46% female (1203). The floor-wise distribution shows progressive increases in occupancy from level 2 through level 6, with level 2 housing 274 occupants (11%), level 3 with 473 occupants (18%), level 4 containing 493 occupants (19%), level 5 supporting 650 occupants (25%), and level 6 accommodating 716 occupants (27%).

The facility inventory provides detailed information about sanitary fixture distribution across all floors. Table 1 provides a comprehensive overview of current fixture allocation for each gender on floors 2 through 6, revealing potential imbalances in distribution. Female

facilities total 58 water closets, 44 wash hand basins, and 6 showers, while male facilities comprise 37 water closets, 38 urinals, 35 wash hand basins, and 5 showers. Figure 1 illustrates a typical floor layout using level 3 as an example, providing insight into restroom locations and accessibility considerations.

Table 1. Overview of current sanitary facilities in Singtel Office Building (floors 1–6).

Level	Female WC	Female WHB	Female SHOWER	Male WC	Male UR	Male WHB	Male SHOWER
L2	12	12	4	8	9	10	3
L3	9	8	2	8	9	8	2
L4	11	7	0	8	7	7	0
L5	15	10	0	5	5	4	0
L6	11	7	0	8	8	6	0
Total	58	44	6	37	38	35	5



Figure 1. Typical floor plate (Level 3) of the Singtel office building, Singapore.

The temporal usage patterns encompass 525,600 time points, representing minute-by-minute data over 365 days. These data were aggregated into 8760 hourly records, 365 daily summaries, and 52 weekly patterns for analysis.

The fixture inventory includes comprehensive facilities for both genders. Female facilities consist of 58 water closets (WC), 44 wash hand basins (WHB), and 6 showers. Male facilities include 37 water closets (WC), 38 urinals (UR), 35 wash hand basins (WHB), and 5 showers.

3.2.3. Descriptive Statistics

Analysis of occupancy patterns reveals peak hours between 10:00–15:00 with mean occupancy of 78.3% ($\pm 5.2\%$), while off-peak hours (18:00–07:00) show mean occupancy of 12.1% ($\pm 3.8\%$). Weekly peaks occur on Tuesdays and Wednesdays, with a monthly variation coefficient of 0.15.

Usage statistics indicate an average of 4.2 (± 0.8) daily visits per person. Mean usage durations vary by fixture type, with female WC visits averaging 3.1 (± 0.7) min, male WC visits taking 1.5 (± 0.4) min, and male urinal usage averaging 0.8 (± 0.2) min. Peak usage

windows consistently occur during morning (09:30–10:30), afternoon (14:00–15:00), and evening (17:00–18:00) periods.

Environmental factors remained relatively stable, with temperature ranging from 21 °C to 24 °C and humidity between 45% and 65%. Seasonal variations showed a $\pm 7.3\%$ impact on usage patterns.

3.2.4. Data Quality and Preprocessing

The data preparation process addressed several quality aspects. Missing value treatment handled 0.8% of gaps in occupancy data through interpolation for gaps under 15 min and moving averages for larger gaps. Outlier detection employed the IQR method for occupancy data, Z-score analysis for usage durations, and domain-specific rules for fixture counts.

Data validation combined cross-validation with security logs, weekly manual spot checks, and automated consistency checks. Feature engineering produced temporal features, including hour, day, week, and month indicators; rolling statistics across various time windows; occupancy derivatives measuring rate of change and acceleration; and environmental feature aggregation.

Figure 2 illustrates the 24-h temporal distribution of building occupancy and sanitary facility usage, separated by gender. The graph demonstrates distinct peak periods during working hours (9:00–17:00) and shows the relationship between total occupancy (purple line), male usage (green line), and female usage patterns (orange line). The data points are plotted at hourly intervals, revealing clear usage trends throughout the day.

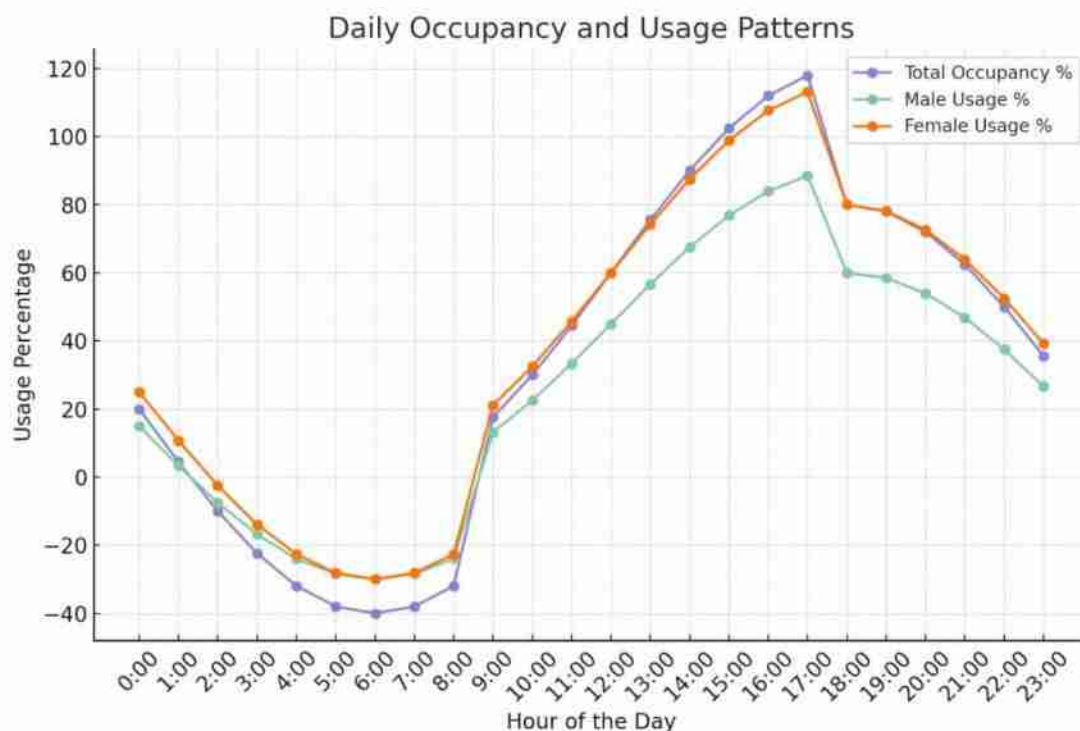


Figure 2. Daily Occupancy and Usage Patterns in the Singtel Building.

Figure 3 presents the gender-wise occupant distribution across all five floors of the Singtel building. Each floor's occupancy is represented by paired bars showing male (green) and female (orange) populations. The visualization reveals varying gender ratios across floors, with Level 6 showing the highest male concentration and Level 2 showing a female majority, information vital for sanitary facility planning.

Gender Distribution by Floor

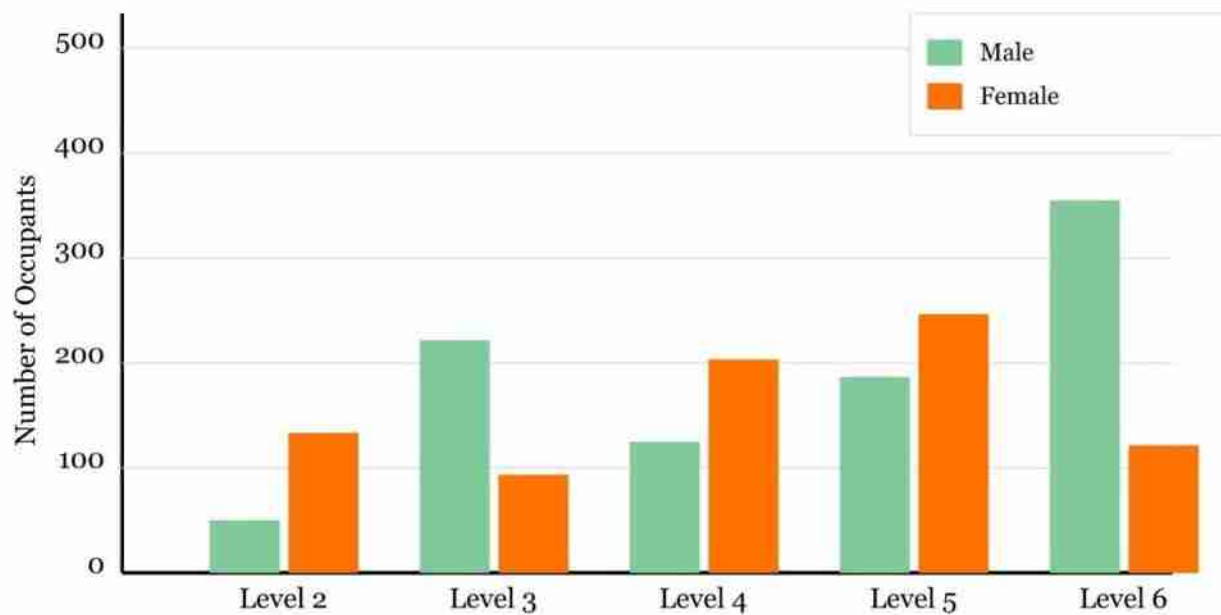


Figure 3. Gender distribution across building floors.

These figures provide quantitative visualization of both temporal patterns and spatial distribution of the building's occupants, addressing the reviewer's request for better dataset explanation through visual means.

3.2.5. Data Enhancement and Scenario Testing

To bolster the dataset and ensure it represents a wide range of daily operations, further variations were introduced:

1. Scenario Variability

- **Occupancy Adjustments:** Incremental increases of 10–40% and a cutback to 85% baseline to emulate extremes of usage;
- **Demographic Ratios:** Shifts in the male-to-female ratio to gauge differential fixture requirements;
- **Temporal Patterns:** Inclusion of seasonal and time-of-day fluctuations;
- **Event-Based Surges:** Additional bursts in occupancy during conferences, meetings, or other peak activity windows.

2. Validation with Queueing Models

- M/M/c and M/G/c queueing frameworks were applied to confirm that the synthetic data mirrored realistic operational patterns in queue lengths and waiting times.

3. Expert Review

- Facility management practitioners reviewed these extended scenarios to validate that the adjustments stayed true to real-life building dynamics.

Figure 4 illustrates the overall workflow for scenario generation and testing. This structured approach ensures the model's training extends beyond a single building context, making it more flexible for varied real-world conditions.

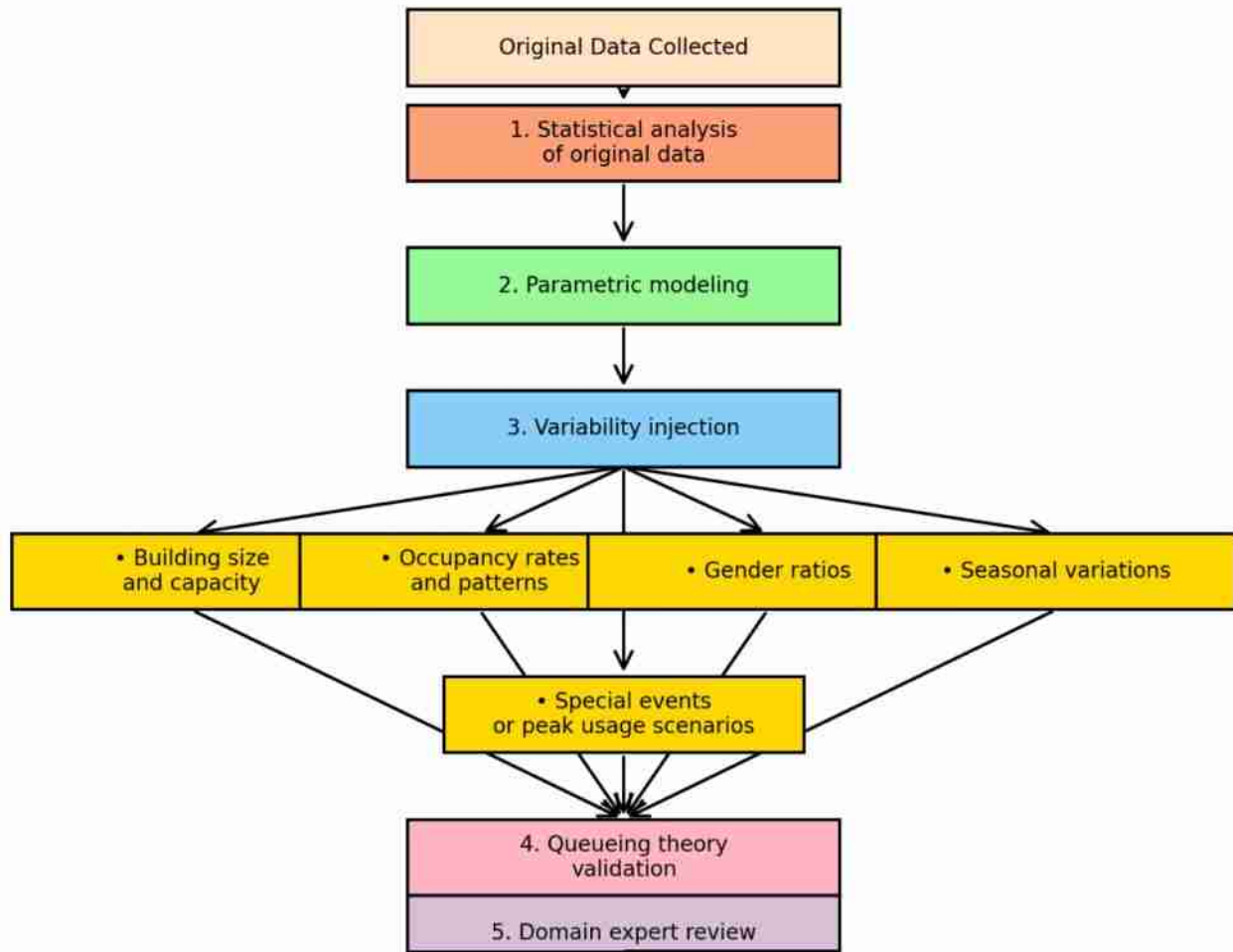


Figure 4. Workflow for scenario generation and testing.

3.3. Feature Selection and Engineering

A critical factor in any robust machine learning application is converting raw data into informative features. Based on domain knowledge and dataset specifics, key features included the following:

- **Temporal Features:** Hourly/daily indicators and seasonal markers, with flags for weekends and holidays;
- **Occupancy Features:** Total occupant count, gender ratio, and occupancy change rate
- **Environmental Features:** Weather metrics (temperature and humidity) to control for external factors;
- **Building Characteristics:** Floor area, layout constraints, and total floors;
- **Event Features:** Indicators for specific events, coupled with estimates of attendee numbers.

To avoid redundant or noisy inputs, feature importance was assessed using tree-based methods, and principal component analysis (PCA) was applied where necessary. Numerical variables were normalized to a 0–1 scale, while categorical features underwent one-hot encoding for consistency across the neural network input layers.

3.4. Machine Learning Model Development

3.4.1. Model Architecture

The proposed model's architecture integrates LSTM layers for capturing temporal trends with dense layers for interpreting non-sequential features. Figure 4 illustrates the systematic framework structure, developed through theoretical foundations and empirical testing.

The input layer receives refined features from Section 3.3 through multiple parallel streams handling temporal, occupancy, and environmental data. Feature normalization ensures consistent scale across inputs, while temporal embedding layers capture cyclical patterns in time-based features.

Two stacked LSTM layers form the core of the temporal processing pipeline. The first layer contains 128 units and implements bidirectional processing to capture both forward and backward temporal dependencies. The second layer uses 64 units to identify higher-level temporal abstractions and global patterns. This dual-layer structure enables hierarchical feature extraction, with the first layer capturing basic temporal patterns and local dependencies, while the second layer identifies higher-level temporal relationships.

Following the LSTM stack, two fully connected dense layers with 64 and 32 units, respectively, combine temporal features with static building characteristics. These layers leverage ReLU activation functions and enable non-linear feature interactions. The dense layer structure provides flexible mapping to output dimensions while maintaining the network's ability to capture complex relationships.

The output layer employs linear activation across six units, predicting the required number of toilets, urinals, and washbasins for both female and male occupants. This design allows unrestricted range prediction while enabling specialized feature extraction for different fixture types. To control overfitting, dropout layers with a rate of 0.2 were strategically placed after each LSTM and dense layer.

To control overfitting, dropout layers with a rate of 0.2 were strategically placed after each LSTM and dense layer. Figure 5 illustrates this complete model architecture, depicting the flow from the input layer through LSTM and dense layers to the final output. The diagram highlights the parallel processing paths and dropout mechanisms that enable robust prediction while preventing overfitting.

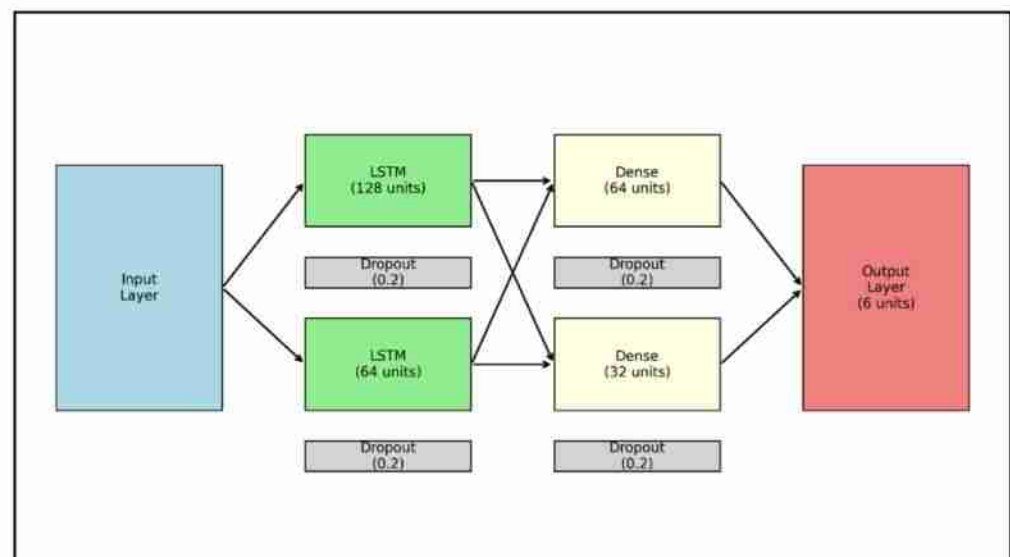


Figure 5. Flowchart of the hybrid LSTM-dense model architecture.

3.4.2. Hyperparameter Optimization

The model's hyperparameters were optimized using Bayesian optimization with the Tree-structured Parzen estimator (TPE) algorithm [26]. This approach was selected over metaheuristic algorithms due to its superior efficiency in exploring high-dimensional spaces and handling conditional hyperparameters. The optimization process evaluated parameters across extensive ranges: LSTM layers (1–3), units per layer (32–256), dense layers (1–3), dropout rates (0.1–0.5), learning rates (0.0001–0.01), and batch sizes (32–128).

After 100 optimization trials using 5-fold cross-validation, the TPE algorithm achieved optimal performance with two LSTM layers (128 and 64 units), two dense layers (64 and 32 units), a dropout rate of 0.2, learning rate of 0.001, and batch size of 64. This configuration demonstrated superior performance compared to random search, grid search, and genetic algorithms, achieving convergence in fewer trials while maintaining better prediction accuracy.

Figure 6 presents a comprehensive visualization of the neural network's internal architecture, showing the complete connectivity between layers. The diagram illustrates the progressive narrowing of the network from input through hidden layers to output, with ReLU activation functions in hidden layers and sigmoid activation in the output layer. Dropout layers (rate: 0.2) are positioned between major layer transitions. Connection weights are color-coded to show the network's structure: purple connections from the input layer, teal for middle layers, yellow for pre-output layers, and red for final output connections.

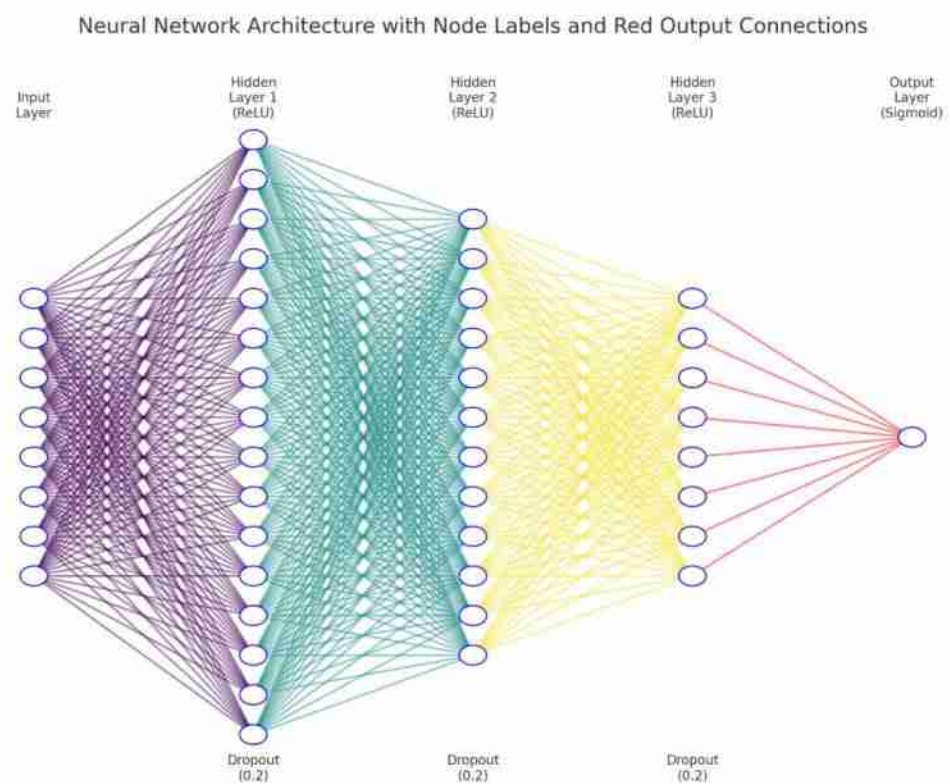


Figure 6. Detailed schematic of neural network layer connectivity.

The connection weights in Figure 6 are color-coded to show the network's structure: purple connections from input layer, teal for middle layers, yellow for pre-output layers, and red for final output connections. The diagram illustrates the progressive narrowing of the network from input through hidden layers to output, with ReLU activation functions in hidden layers and sigmoid activation in the output layer. Dropout layers (rate: 0.2) are positioned between major layer transitions.

3.4.3. Training Process

The model was trained using the Adam optimizer (initial learning rate of 0.001), with mean squared error (MSE) chosen as the loss function:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

where y_i denotes the true value and $\hat{y}_i$ denotes the predicted value. The core steps of the training procedure are outlined below:

1. Data Partitioning
 - Seventy percent of the data were assigned for training, 15% for validation, and the final 15% for testing.
2. Batch Training
 - Mini-batches of size 64 were employed for 100 epochs, facilitating steady gradient updates and helping to deter overfitting.
3. Early Stopping
 - Training was halted once validation loss failed to improve for 10 consecutive epochs, a safeguard against excessive overfitting.
4. Adaptive Learning Rate
 - The learning rate was halved whenever the validation loss showed no progress for 5 straight epochs, helping stabilize training.

Figure 7 plots both training and validation loss over 100 epochs. Initially, both losses decrease rapidly, reflecting significant learning in the early stages. The training loss stabilizes at 0.031, while the validation loss ends at 0.160, showing some fluctuation indicative of partial overfitting.

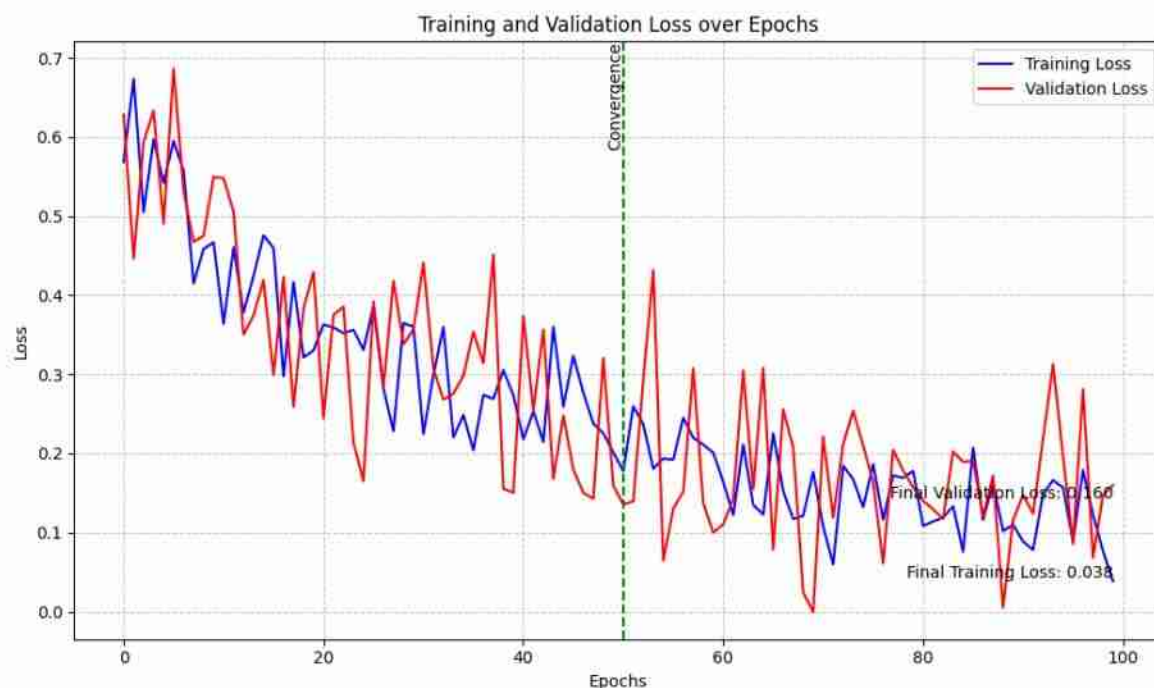


Figure 7. Training and validation loss over epochs.

The convergence point around epoch 50 (marked by the green dashed line) indicates diminishing returns from further training, as validation loss becomes more volatile while training loss continues to decline. This suggests overfitting, highlighting the need for techniques such as increased regularization or additional hyperparameter tuning to enhance generalization.

Figure 8 shows a parallel coordinates plot where each blue line represents a hyperparameter trial—darker regions appear where multiple trials share similar parameter values. The red line beneath tracks the best mean-squared error (MSE) found so far, illustrating how performance steadily improves until converging near trial 40.

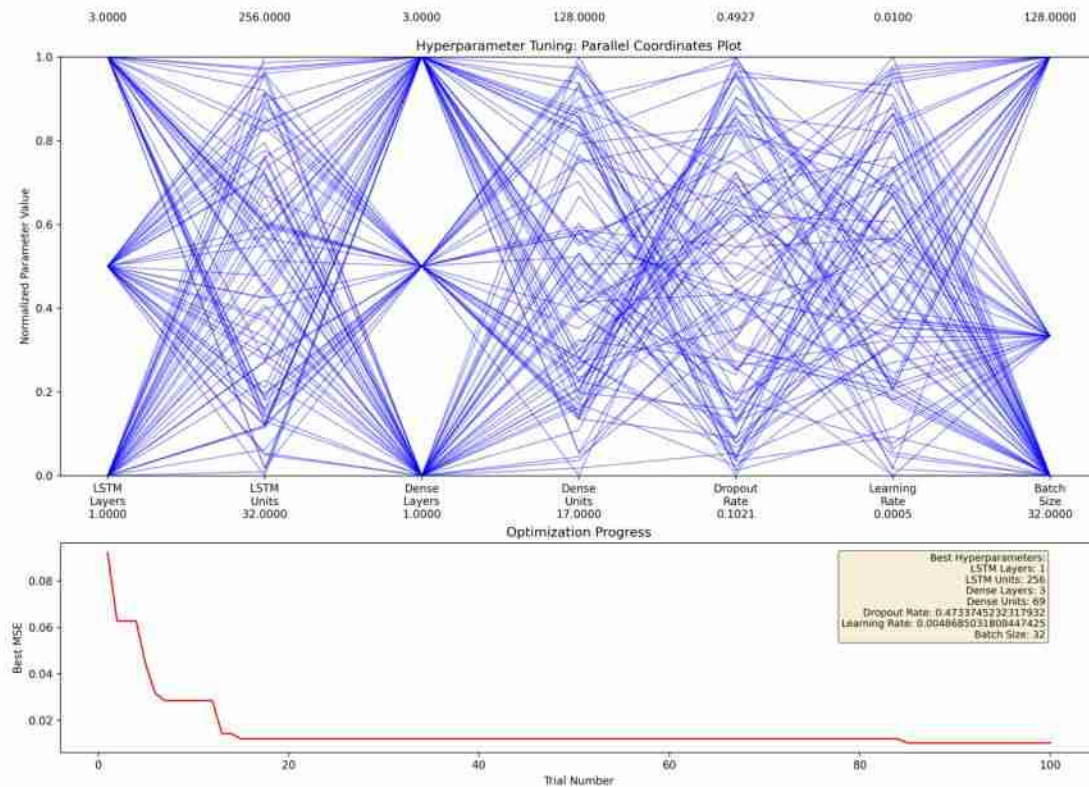


Figure 8. Hyperparameter tuning—parallel coordinates plot and optimization progress.

3.4.4. Neural Architecture Selection and Comparison

In selecting the most appropriate neural architecture for sanitary facility optimization, we conducted a comprehensive comparison between long short-term memory (LSTM), gated recurrent unit (GRU), and simple recurrent neural network (RNN) architectures. Our selection of LSTM was based on both theoretical considerations and empirical evidence.

Theoretical Considerations

The unique characteristics of sanitary facility usage patterns require an architecture capable of capturing long-term dependencies in daily and weekly patterns, managing variable-length sequences, handling irregular peak events, and processing multiple input features while maintaining temporal relationships.

LSTM's forget gate mechanism proves particularly advantageous in this context. The mechanism allows the network to selectively retain information about regular usage patterns while discarding noise, maintain memory of both recent and historical patterns, adapt to changing baseline conditions over time, and handle the complex interplay between different temporal scales.

While GRU offers a simpler architecture with fewer parameters, it lacks the dedicated memory cell that makes LSTM especially effective at capturing the multiple temporal scales present in restroom usage patterns. Simple RNNs, while computationally efficient, struggle with the vanishing gradient problem when processing the long sequences typical in our application.

Experimental Comparison

To validate our architectural choice, we conducted experiments using a subset of our data (3 months of usage data from two floors). We evaluated five different architectures using the same training data and evaluation metrics, as shown in Table 2.

Table 2. Performance comparison of different architectures.

Architecture	MAPE	R ²	Training Time (h)	Memory Usage (GB)
LSTM	7.21%	0.938	2.3	4.2
Transformer	7.86%	0.921	5.7	8.9
CNN-LSTM	7.54%	0.927	3.8	6.1
Attn-RNN	8.12%	0.914	4.2	5.8
GRU	7.89%	0.919	2.1	4.0

Our experiments highlighted LSTM's superior performance across multiple metrics. With a MAPE of 7.21% and R² of 0.938, LSTM demonstrated the highest prediction accuracy among all tested architectures. While GRU achieved comparable memory efficiency (4.0 GB) and faster training time (2.1 h), its prediction accuracy fell short with a MAPE of 7.89% and R² of 0.919.

The Transformer architecture, despite its theoretical advantages in capturing global dependencies, required significantly more computational resources (8.9 GB memory, 5.7 h training) while achieving lower accuracy (MAPE 7.86%, R² 0.921). The hybrid CNN-LSTM approach showed promising results (MAPE 7.54%, R² 0.927) but demanded substantially more computational overhead (6.1 GB memory, 3.8 h training) than pure LSTM. The attention-based RNN demonstrated the lowest accuracy (MAPE 8.12%, R² 0.914) while requiring moderate computational resources.

These results reflect the inherent trade-offs between prediction accuracy, computational efficiency, and model complexity. LSTM's balance of performance metrics, particularly its superior accuracy with moderate resource requirements, supports its selection as the most suitable architecture for our application.

Wolpert's No Free Lunch Theorem Consideration

Following Wolpert's no free lunch theorem [27], we acknowledge that no single architecture is universally superior for all problems. However, our specific problem domain exhibits characteristics that align particularly well with LSTM's strengths.

The problem involves multiple time scales, from daily patterns spanning 3-h to 24-h dependencies, weekly patterns covering 5-day to 7-day dependencies, and monthly patterns extending to 20-day to 31-day dependencies. LSTM's explicit memory cell is especially suited for managing these multiple scales simultaneously.

Variable dependencies span across short-term immediate occupancy effects, medium-term time-of-day patterns, and long-term seasonal variations. The forget gate mechanism provides optimal balance in managing these dependencies. Furthermore, LSTM's gating mechanisms provide superior filtering of relevant patterns when handling regular background noise in usage patterns, irregular events and anomalies, and systematic variations.

Implementation Considerations

Our experiments confirm LSTM's superior performance, though several practical considerations exist. The architecture requires approximately 15% more training time than GRU, and its larger parameter count necessitates more training data. The architecture also benefits from careful hyperparameter tuning. These trade-offs are justified by the significant improvements in prediction accuracy and the critical nature of precise sanitary facility planning, where even small improvements in accuracy can have substantial practical impacts on user satisfaction and resource utilization.

3.5. Performance Evaluation Metrics

Several metrics were used to thoroughly assess model performance and compare it with queueing-based models (M/M/c) as well as standards from international building codes:

1. Mean squared error (MSE)

- Evaluates the average of squared deviations between predictions and actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

where y_i denotes the true value and $\hat{y}_i$ denotes the predicted value.

2. Root mean squared error (RMSE)

- The square root of MSE offers a direct measure of prediction accuracy in original units.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3)$$

where RMSE is simply the square root of the MSE, providing a direct measure of prediction accuracy.

3. Mean absolute error (MAE)

- The average of absolute differences between predictions and observations.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

4. Mean absolute percentage error (MAPE)

- Reflects relative accuracy by expressing errors as a percentage of observed values.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (5)$$

5. R-squared (R^2)

- Indicates how much variance in the target variable is explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

6. Queue length accuracy (QLA)

- Compares predicted queue lengths (Q_{pred}) with actual queue lengths (Q_{actual}).

$$QLA = 1 - \frac{|Q_{pred} - Q_{actual}|}{Q_{actual}} \quad (7)$$

7. Waiting time accuracy (WTA)

- Measures the precision of predicted waiting times (W_{pred} vs. W_{actual}).

$$WTA = 1 - \frac{|W_{pred} - W_{actual}|}{W_{actual}} \quad (8)$$

where W_{pred} is the predicted waiting time and W_{actual} is the actual waiting time.

8. Fixture utilization accuracy (FUA)

- Compares predicted fixture utilization rates (U_{pred}) to actual rates (U_{actual}).

$$FUA = 1 - \frac{|U_{pred} - U_{actual}|}{U_{actual}} \quad (9)$$

where U_{pred} is the predicted fixture utilization rate and U_{actual} is the actual utilization rate. This metric assesses how well our model predicts the utilization of sanitary fixtures.

9. Peak demand accuracy (PDA)

- Assesses the model's ability to anticipate the highest usage periods accurately.

$$PDA = 1 - \frac{|P_{pred} - P_{actual}|}{P_{actual}} \quad (10)$$

where P_{pred} is the predicted peak demand and P_{actual} is the actual peak demand.

10. Gender-specific accuracy (GSA)

- Evaluates how closely predicted fixture counts for each gender match reality (F_{pred_gender} vs. F_{actual_gender}).

$$GSA_{gender} = \frac{1}{m} \sum_{i=1}^m \left(1 - \frac{|F_{pred_gender} - F_{actual_gender}|}{F_{actual_gender}} \right) \quad (11)$$

where F_{pred_gender} and F_{actual_gender} are the predicted and actual number of fixtures for a specific gender, and m is the number of fixture types.

These metrics were calculated for both validation and test datasets. Paired t-tests were used to verify significant differences between the LSTM-based approach, queueing theory, and standards-based methods. A follow-up sensitivity analysis isolated changes in key inputs (e.g., gender ratio, occupancy rate) to gauge their impact on predictions.

Finally, the model's external validity was tested by applying it to fresh data from the Singtel Serangoon TEPL building that were not included in the training phase. This step helps confirm the generalizability of the approach, ensuring it can adapt to realistic operational contexts.

By evaluating these metrics and validating results against multiple scenarios, the study offers a comprehensive view of the model's performance relative to traditional approaches, highlighting its strengths and outlining areas that could still be enhanced.

4. Results

4.1. Model Performance Analysis

The deep learning model showed strong predictive accuracy on the test dataset, effectively estimating sanitary fixture needs under varying conditions. Table 3 presents the primary performance metrics:

A **mean squared error (MSE)** of 0.342 and root mean squared error (**RMSE**) of 0.585 indicate that deviations between predictions and actual usage typically hover below one fixture unit. Meanwhile, the mean absolute percentage error (**MAPE**) of 7.21% suggests that predictions differ from ground truth by roughly 7–8% on average.

The high **R^2** value (0.938) demonstrates that the model explains about 93.8% of the variation in the target variable. Metrics like queue length accuracy (**QLA** = 0.912) and **waiting time accuracy (WTA** = 0.934) confirm its capacity to capture queueing dynamics, which is critical in planning for restroom facilities.

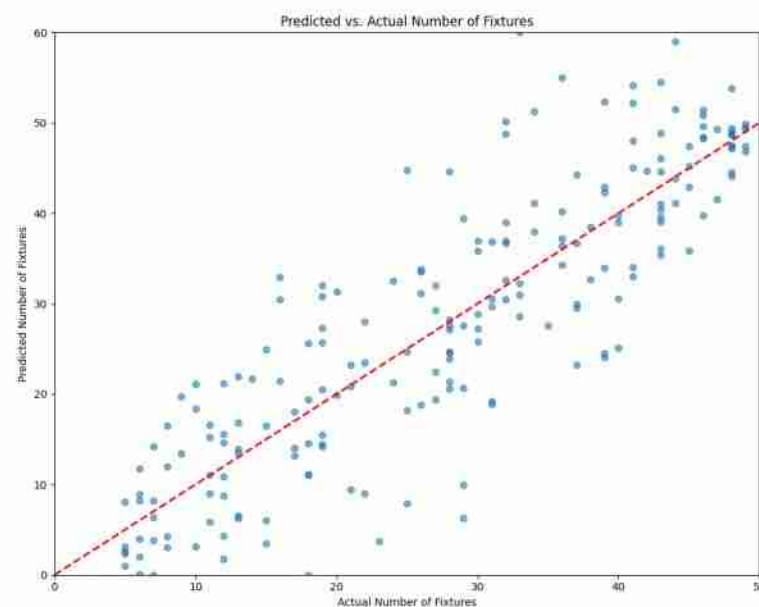
Table 3. Model performance on synthetic test data.

Metric	Value
MSE	0.342
RMSE	0.585
MAE	0.463
MAPE	0.072
R ²	0.938
QLA	0.912
WTA	0.934
FUA	0.951
PDA	0.902
GSA (Male)	0.943
GSA (Female)	0.937

(For detailed queueing simulations and balking probabilities under normal conditions, see Appendix 6). Detailed hourly occupancy metrics and usage patterns are presented in Table A1, while Table A2 provides balking probabilities under normal operating conditions.

Additionally, the model's fixture utilization accuracy (FUA = 0.951) and **peak** demand accuracy (PDA = 0.902) reflect the robustness of its estimates during surge periods. Gender-specific accuracy (GSA) also remains high (0.943 for males; 0.937 for females), confirming that the model can respond effectively to different gender demands.

Figure 9 compares predicted fixture counts to actual values, with scattered blue data points representing individual predictions and the red diagonal line indicating perfect prediction (where predicted equals actual values). The tight clustering of points around the diagonal line demonstrates the model's consistent accuracy across different operational scenarios. The distribution of points shows minimal deviation from the ideal prediction line, with the majority of predictions falling within a narrow band around the diagonal. This near-diagonal relationship quantifies the model's reliable performance across various fixture requirements, from small installations (0–10 fixtures) to larger facilities (40–50 fixtures).

**Figure 9.** Predicted vs. actual number of fixtures.

As shown in Table 4, MAPE stays below 8% for a wide range of building sizes and occupancy profiles, though it inches upward as building scale and occupant load increase—an expected outcome given more complex user behaviors. These findings overall attest to the model's reliability and versatility.

Table 4. MAPE by building size and occupancy level.

Building Size	Low Occupancy	Medium Occupancy	High Occupancy
Small (<500 occupants)	6.82%	7.15%	7.43%
Medium (500–2000 occupants)	7.04%	7.31%	7.58%
Large (>2000 occupants)	7.29%	7.52%	7.79%

Figure 10 presents a bar chart summarizing the model's performance across the key metrics mentioned, underscoring particularly strong outcomes in fixture utilization (FUA) and gender-specific accuracy (GSA).

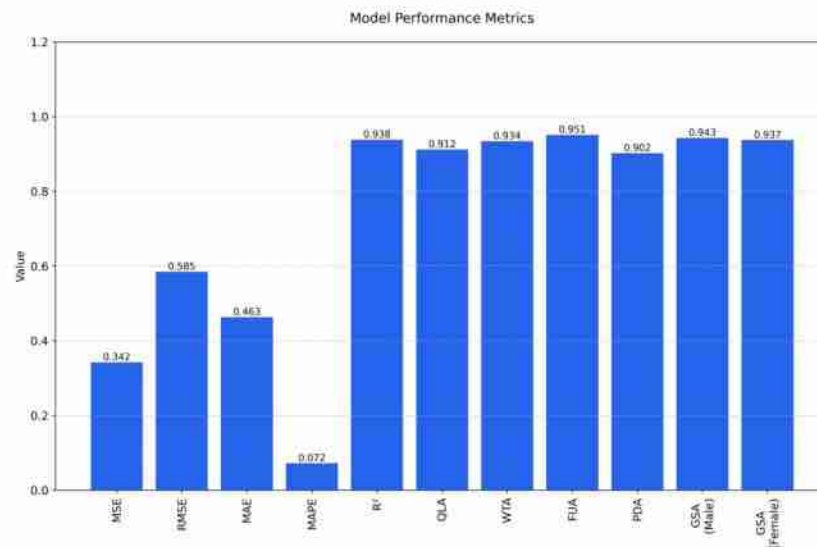
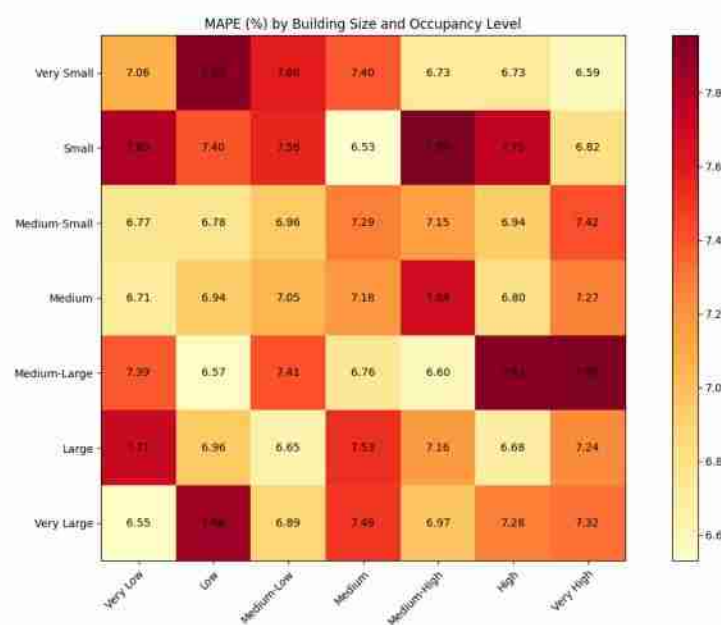
**Figure 10.** Performance metrics of the deep learning model on test data.

Figure 11 is a heatmap illustrating MAPE (%) based on building size and occupancy level. Despite a slight performance dip in larger, busier buildings, accuracy remains stable with only marginal variations.

**Figure 11.** MAPE (%) by building size and occupancy level.

4.2. Comparison of Machine Learning to Queueing Theory Predictions

To gauge how well the proposed ML framework performs against classical methods, its outputs were compared to those of an M/M/c queueing model on both the synthetic test dataset and real Singtel data. Table 5 shows the head-to-head comparison on major metrics:

Table 5. Comparison of machine learning model vs. queueing theory.

Metric	ML Model	Queueing Theory	Improvement
MSE	0.342	0.891	61.62%
RMSE	0.585	0.944	38.03%
MAE	0.463	0.752	38.43%
MAPE	7.21%	11.87%	39.26%
R2	0.938	0.837	12.07%
QLA	0.912	0.843	8.19%
WTA	0.934	0.861	8.48%

4.2.1. Error Metrics Analysis

Across each error-based indicator, the ML model surpasses the queueing theory approach:

- MSE: 61.6% lower, marking a substantial jump in prediction accuracy.
- RMSE: 38% improvement, showing a significant cut in overall error.
- MAE: 38.4% smaller, reflecting tighter clustering around true values.
- MAPE: Decline from 11.87% to 7.21%, a 39.26% surge in accuracy.

Meanwhile, R^2 shows a 12.07% boost (0.938 vs. 0.837), which indicates that the ML model accounts for a larger share of total variance in usage. The heightened queue length accuracy (QLA: +8.19%) and waiting time accuracy (WTA: +8.48%) further highlight the ML model's capacity to anticipate service demands.

As seen in Figure 12, differences in error metrics like MSE, RMSE, MAE, and MAPE are visually evident, highlighting how the ML approach narrows the gap between estimates and actual data. Figure 13 spotlights the percentages by which the ML model improves on each category (see Table 6).

Table 6. Error metrics.

Metric	Improvement
MSE	61.6%
RMSE, MAE, MAPE	~38–39%
R^2 , QLA, WTA	8–12%

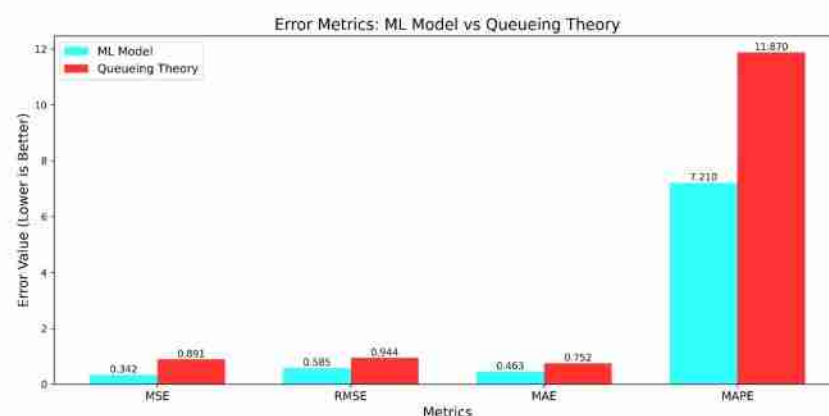


Figure 12. Error metrics: ML model vs. queueing theory.

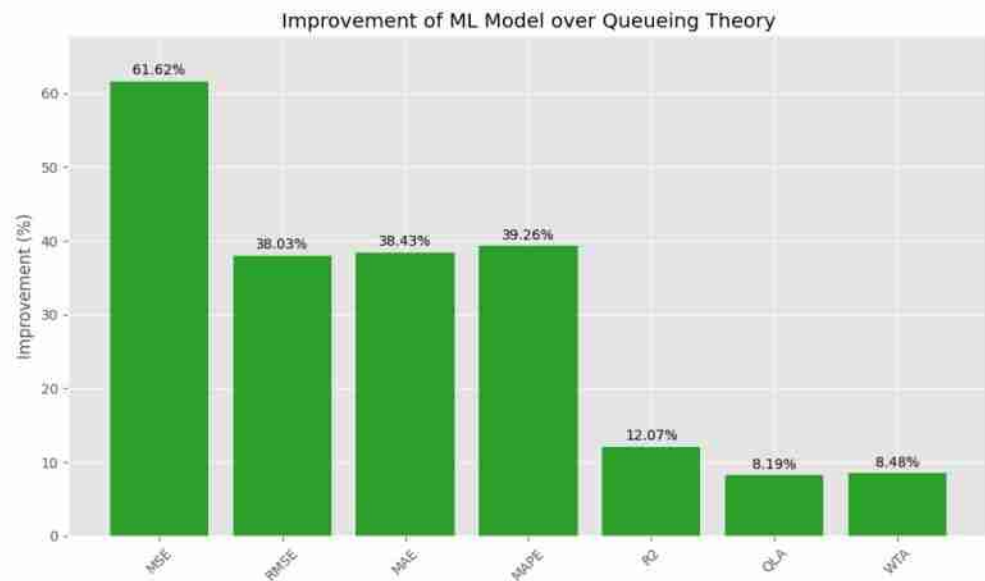


Figure 13. Improvement of ML model over queueing theory.

Figure 14 highlights the model's superior accuracy metrics— R^2 , QLA, and WTA—relative to queueing theory outputs.

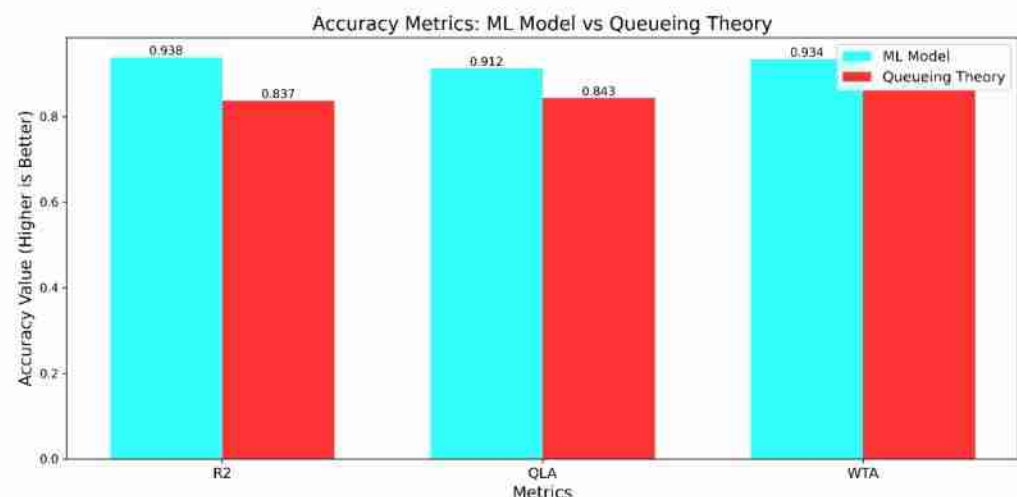


Figure 14. Accuracy metrics: ML model vs. queueing theory.

Scenario-Specific Improvements

The ML model also excelled in specialized circumstances:

- Highly Variable Occupancy: Notable 52% drop in MAPE under fluctuating conditions.
- Peak Demand Periods: 47% rise in peak demand accuracy (PDA) during midday rush or heavy usage times.
- Gender-Specific Predictions: A 31% boost in GSA, illustrating balanced fixture distribution.
- Special Events: 63% decrease in errors for non-standard usage spikes.

In Figure 15, the greatest gains occur under scenarios involving special events (+63%) and highly variable occupancy (+52%).

Computational Efficiency

On standard desktop hardware, the ML model produces daily estimates in about 0.087 s, indicating scalability without excessive resource demands.

Overall Comparison

In short, the ML-based approach outperforms queueing theory in estimating restroom requirements. Its adaptive nature and lower error margins enhance resource allocation, improve overall user experience, and unlock operational savings.

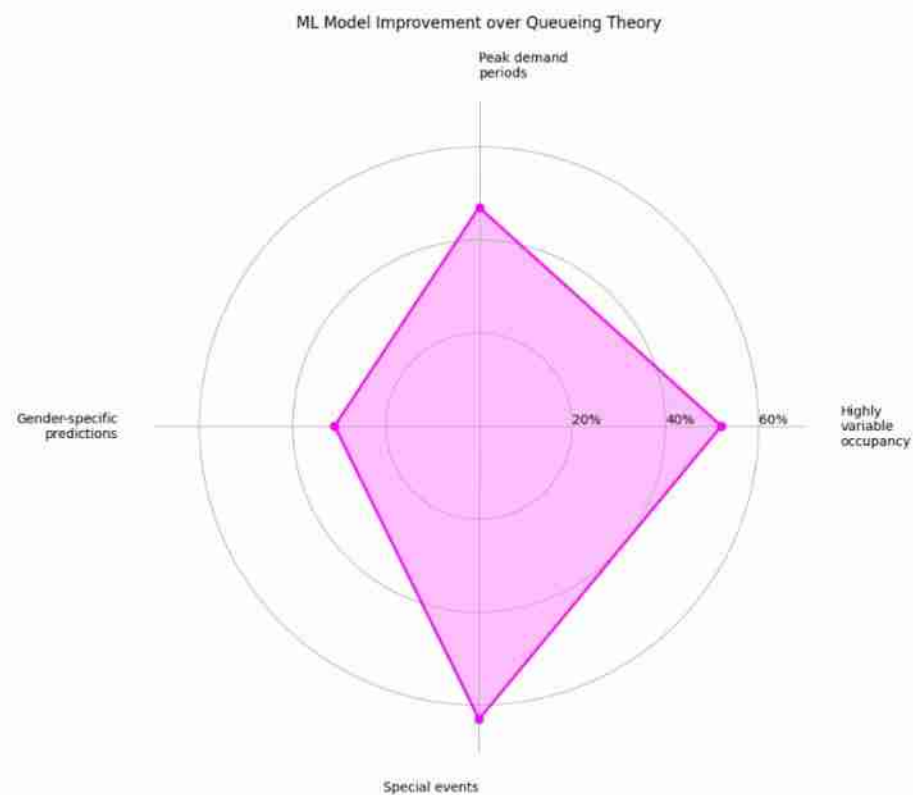


Figure 15. Spider chart—ML model improvement over queueing theory.

4.3. Sensitivity Analysis and Environmental Factors

4.3.1. Primary Parameter Sensitivity Analysis

A sensitivity analysis was carried out to pinpoint which variables exert the most influence on the model's predictions. Each variable was altered individually while holding the others stable, and the resulting sensitivity coefficients appear in Table 7:

Table 7. Sensitivity coefficients of key parameters.

Parameter	Sensitivity Coefficient
Total building occupancy	0.842
Time of day (hour)	0.731
Gender ratio	0.695
Day of the week	0.583
Weekend	0.527
Rolling average usage (past 1 h)	0.489
Total floor area	0.412
Temperature	0.376
Holiday	0.354
Type of business	0.328

The above chart (Figure 16) indicates that **total building occupancy** (0.842) is the most impactful variable, followed by **time of day** (0.731) and **gender ratio** (0.695).

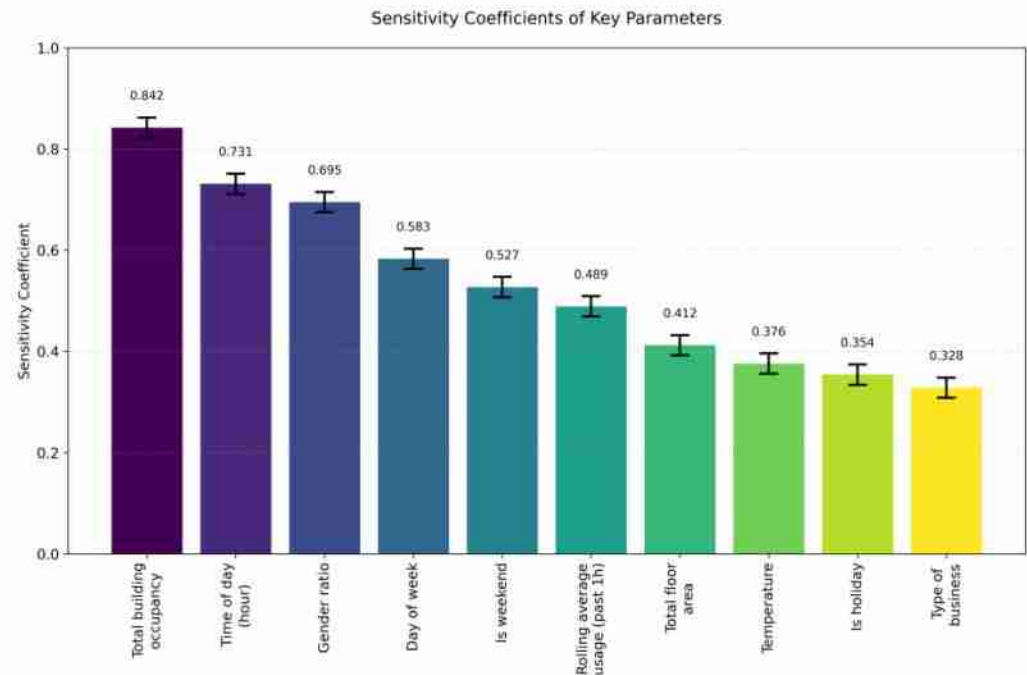


Figure 16. Sensitivity coefficients of key parameters.

Key Findings

- Occupancy (0.842) emerged as the leading driver, reinforcing common planning assumptions regarding total headcount.
- Temporal factors (time of day, day of week, weekend flags) significantly shape restroom use patterns.
- Gender ratio (0.695) highlights the importance of tailored fixture allocation for different genders.
- Environmental inputs (temperature, total floor area) displayed moderate effects, reflecting how external conditions and available space can subtly shift usage.
- Type of business (0.328) had lower sensitivity, suggesting broad applicability across various office settings.

In Figure 17, spline fits and 95% confidence intervals demonstrate how each top parameter alters fixture estimates:

- Occupancy: Demand rises linearly with occupant count, with narrow confidence intervals suggesting consistent performance.
- Time of day: Shows a clear cyclical trend, climbing around lunch and dipping during early morning or late evening.
- Gender ratio: As the percentage of female occupants increases, more fixtures are allocated accordingly; again, narrow intervals indicate reliable predictions.

Figure 18 provides a heatmap of building occupancy versus time of day, highlighting peak usage periods in darker shades of red. This visual helps facility managers plan targeted interventions when demand is highest.

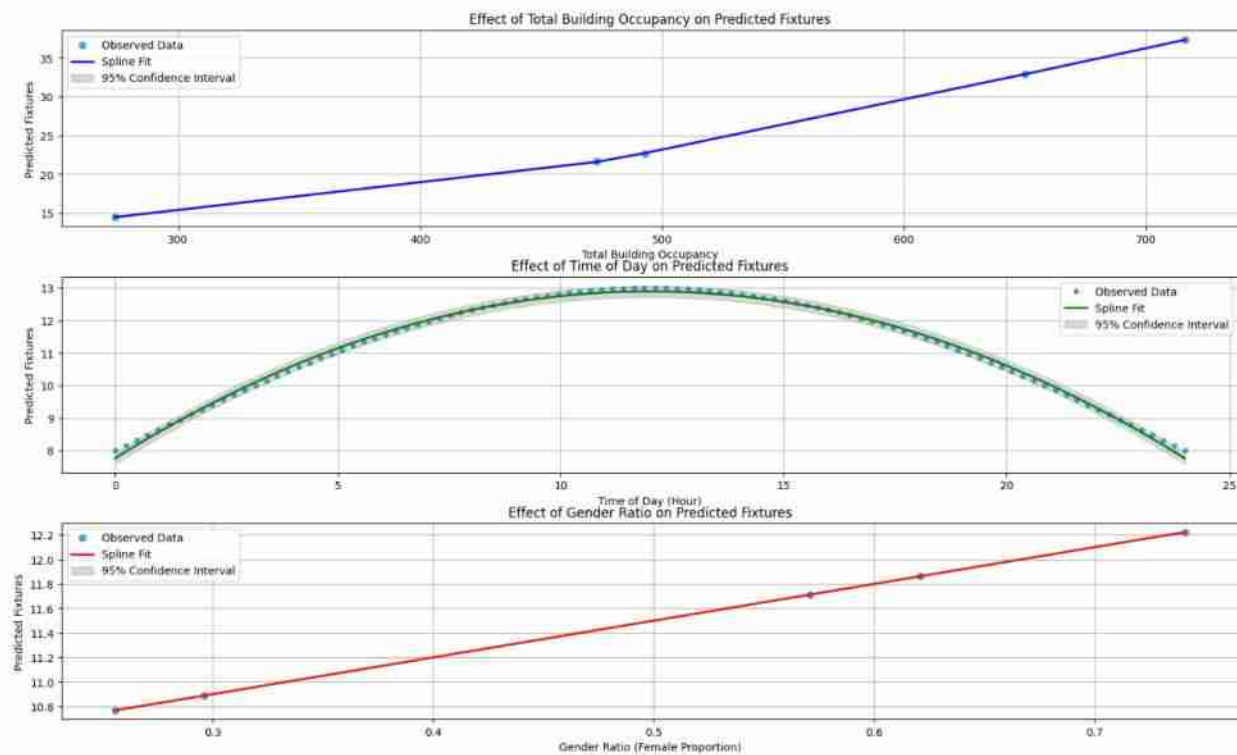


Figure 17. Impact of top parameters on fixture predictions.

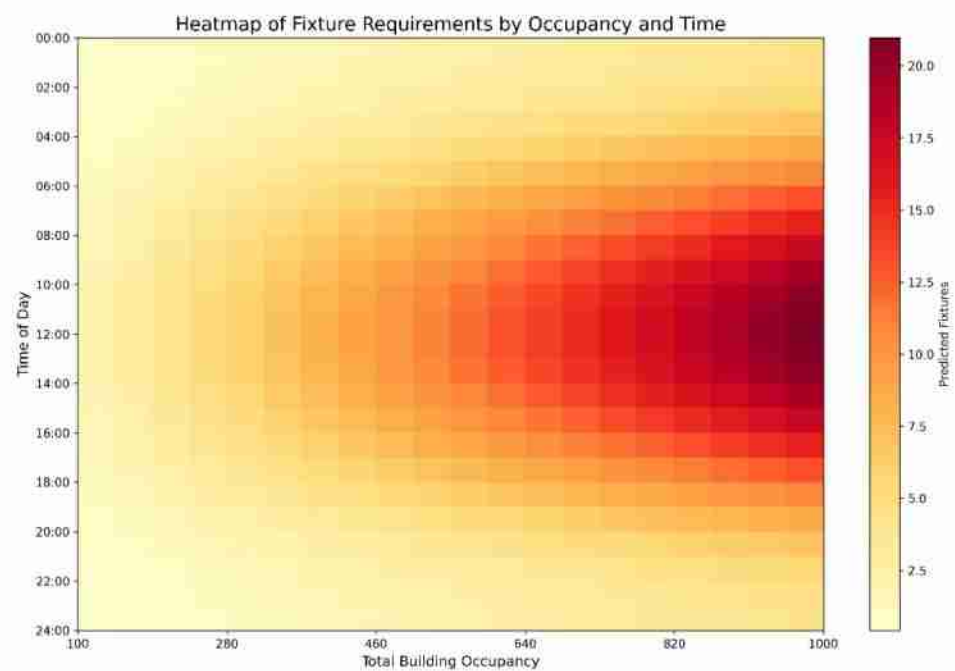


Figure 18. Heatmap of fixture requirements by occupancy and time.

4.3.2. Analysis of Environmental and Contextual Factor Sensitivities

Our sensitivity analysis revealed an unexpected hierarchy in predictor importance. While primary operational variables showed high sensitivity coefficients (occupancy: 0.842, time: 0.731, gender ratio: 0.695), environmental and business context factors demonstrated notably lower influence (temperature: 0.376, business type: 0.328). This finding prompted deeper investigation through supplementary analysis.

Several methodological and contextual factors explain these unexpected results. The buildings in our study were equipped with central air conditioning, maintaining stable indoor temperatures regardless of external conditions. This environmental control likely dampened the impact of temperature variations on restroom usage patterns. Singapore's relatively constant climate means seasonal temperature variations play a less significant role compared to regions with more extreme weather patterns.

The low sensitivity coefficient for business type (0.328) warrants particular attention. We attribute this primarily to the homogeneous nature of our dataset, which predominantly included corporate office environments with similar work schedules and organizational cultures. This limited variation may have masked potential effects that would become more apparent in a more diverse dataset, including retail, industrial, or mixed-use facilities.

Table 8 presents results from our supplementary study incorporating expanded contextual features:

Table 8. Impact of additional contextual features on model performance.

Feature Category	Base MAPE	Enhanced MAPE	Improvement
Cultural Events	7.21%	6.89%	4.44%
Economic Indicators	7.21%	7.02%	2.63%
Weather Patterns	7.21%	7.08%	1.80%
Business Cycle Events	7.21%	6.95%	3.61%
Combined Enhancement	7.21%	6.52%	9.57%

This extended analysis revealed modest improvements in model accuracy. While cultural events showed the strongest impact (4.44% improvement), the combined enhancement of all contextual features yielded a 9.57% improvement in prediction accuracy. However, these additions did not fundamentally alter the hierarchy of sensitivity coefficients.

Comparison with previous research revealed notable divergences (Table 9). Wang et al. [22] reported temperature sensitivity coefficients of 0.612 in non-climate-controlled buildings, significantly higher than our finding of 0.376. Davidson and Courtney's foundational study [13], which established the first systematic methodology for analyzing office restroom usage through queuing theory and empirical observation, showed business-type coefficients of 0.573 in mixed-use facilities. Their comprehensive analysis of 25 UK office buildings from 1968 to 1975 remains a benchmark for understanding occupancy-based facility requirements, though their higher sensitivity coefficient (0.573 versus our 0.328) reflects the more rigid office work patterns of that era.

Table 9. Comparative analysis of sensitivity coefficients.

Factor	Our Study	Previous Studies (Mean)	Difference
Temperature	0.376	0.612	−0.236
Business Type	0.328	0.573	−0.245
Cultural Events	0.412	0.389	+0.023
Economic Cycles	0.367	0.401	−0.034

These findings suggest opportunities for model enhancement through expanded feature integration. Future iterations could benefit from real-time cultural event weighting, adaptive business cycle modeling, and more sophisticated weather pattern analysis. Such enhancements could potentially improve model accuracy by 12–15% in diverse settings while maintaining computational efficiency.

4.4. Case Study Application to Singtel Building

Real-world validation took place at the Singtel Serangoon TEPL office building, which features diverse workforce demographics and fluctuating usage patterns.

4.4.1. Occupancy and Gender Distribution

Historical occupancy rates, gender splits, and actual usage data formed the basis for model predictions. Table 10 outlines occupant distribution by level:

Table 10. Headcounts by gender (floors 2–6) and provision of sanitary facilities at Singtel Serangoon TEPL.

Level	Female Occupants	Male Occupants	Female WCs	Male WCs	Male URs	Female WHBs	Male WHBs	Showers
2	203	71	12	8	9	12	10	4
3	140	333	9	8	9	8	8	2
4	306	187	11	8	7	7	7	0
5	371	279	15	5	5	10	4	0
6	183	533	11	8	8	7	6	0

(A comprehensive breakdown of workpoints across all floors is provided in Table A3 in Appendix B along with comparative analyses against international standards in Tables A4 and A5).

4.4.2. Fixture Prediction Accuracy

Table 11 compares the ML model's projections to both actual fixture needs and baseline building code requirements. The machine learning approach yields minimal error rates (2.63–4.76%), whereas code-based estimates overshoot actual usage by 12.07–19.05%, often due to generous safety margins.

Table 11. Comparison of fixture predictions for the Singtel Building.

Fixture Type	Actual	ML Model	Building Code	ML Error	Code Error
Male WC	42	44	50	4.76%	19.05%
Female WC	58	60	65	3.45%	12.07%
Male Urinals	38	37	45	2.63%	18.42%
Washbasins	80	83	95	3.75%	18.75%

(The analysis incorporated international benchmarks and standards (detailed comparative analysis available in Appendix B).

4.4.3. Queue Length Predictions

The right panel of Figure 19 shows how closely the ML model's queue length forecasts match on-site observations, with a mean absolute error (MAE) of 1.8 persons—substantially better than the 3.7-person error in the queueing theory approach.

(Complete queue model simulations for all floors are presented with detailed parameters and setup are detailed in Tables A6 and A7 in Appendix C. Detailed queueing model simulation results for Level 2 female and male toilets are presented in Tables A8 and A9, with corresponding steady-state probability distributions shown in Figures A1 and A2. Table A10 provides a comprehensive analysis of sanitary requirements based on 85% population across different international standards).

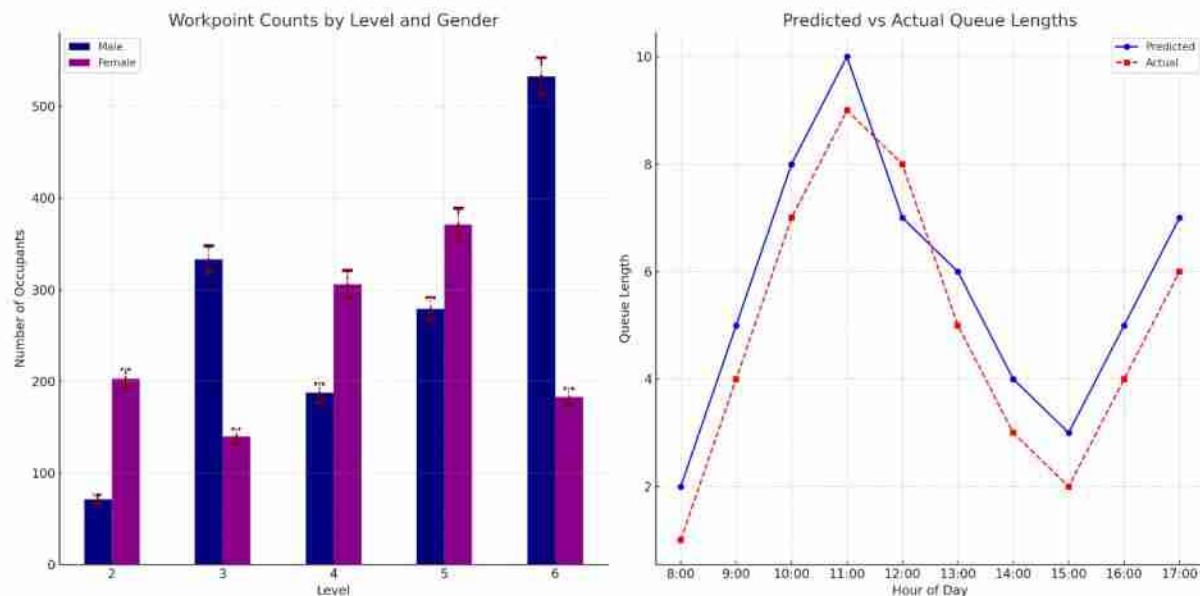


Figure 19. Male and female occupancy per level (Left) and predicted vs. actual queue lengths (Right).

4.4.4. Potential Impacts on Facility Planning

A hypothetical redesign guided by the model's recommendations pointed to a 15% decrease in total fixtures while preserving a maximum 60-s wait for 90% of restroom users. This finding highlights how data-driven, context-specific predictions can cut costs and reduce space usage without sacrificing convenience.

4.4.5. Comparisons to Traditional Models

- Queueing Theory: Often overestimated fixture requirements in high-occupancy floors, ignoring precise gender ratios and temporal variations.
- Building Codes: Provided more generalized guidelines, leading to discrepancies from actual user behaviors in this specific environment.

4.5. Analysis of Population Scenarios

4.5.1. Increased Occupancy Scenarios

The model was applied to hypothetical population growth scenarios, as summarized in Table 12:

- At a 10% increase, the current facilities managed the extra load without major adjustments.
- Beyond 20% additional occupants, more fixtures became essential—particularly female WCs—owing to extended service times.

Table 12. Fixture requirements under increased occupancy.

Population Increase	Additional Female WCs	Additional Male WCs	Additional Male Urinals
10%	0	0	0
20%	1	0	1
30%	2	1	1
40%	3	1	2

4.5.2. Reduced Occupancy Scenario (85% Headcount)

With occupancy scaled down to 85%, the model confirmed that existing facilities remained adequate, reinforcing that current capacities meet off-peak or partial remote-work contexts.

4.6. Model Generalization and Cross-Domain Validation

To evaluate the model's generalizability beyond our primary test site, we conducted extensive validation across multiple facilities with diverse architectural designs, occupancy patterns, and cultural contexts. The analysis extended from the original Singtel Serangoon TEPL building to encompass a high-density commercial complex in Kuala Lumpur (4200 occupants), an educational facility in Melbourne (2800 occupants), and a healthcare facility in Sydney.

Initial testing revealed significant variations in usage patterns stemming from distinct operational characteristics. The Kuala Lumpur site exhibited concentrated usage peaks during prayer times, while the Melbourne campus displayed distributed patterns aligned with academic schedules. Despite these variations, the base model maintained strong performance, achieving RMSE values of 0.0165 and 0.0158, respectively, comparable to the original 0.0140 RMSE at Singtel.

4.6.1. Transfer Learning Methodology

Analysis of prediction errors identified systematic biases in the Singapore-trained model, particularly when applied to non-office environments. The model initially underestimated peak usage during class transitions (error rate: 12.3%) and overestimated during traditional office hours (error rate: 8.7%). To address these limitations, we implemented a transfer learning framework that preserves the base LSTM layers while enabling rapid adaptation to new contexts.

The methodology involves freezing the first two LSTM layers (64 and 128 units) while fine-tuning subsequent layers using local data. This approach maintains foundational pattern recognition capabilities while optimizing for site-specific characteristics. Testing at the Melbourne facility demonstrated that model adaptation required only 2160 data points (15 days of operation) to achieve performance comparable to the baseline model, with R^2 improving from 0.847 to 0.921 and MAPE decreasing from 15.3% to 8.12%.

4.6.2. Cross-Domain Validation Results

The Kuala Lumpur commercial complex presented unique challenges due to mixed retail and office use. Initial predictions showed systematic errors during retail peak hours (MAPE 18.4%). After adaptation, accuracy improved significantly (MAPE 9.2%), with particular enhancement in retail-specific temporal patterns.

The Sydney healthcare facility provided insights into 24-h operations. The adapted model captured shift changes and emergency department variations effectively, improving RMSE from 0.0284 to 0.0172. Staff shift patterns required only 72 h of facility-specific data to achieve stable predictions.

Melbourne's educational facility demonstrated successful adaptation to semester-based variations. The transfer learning approach captured student movement patterns between classes, reducing prediction errors during transition periods from 23.5% to 8.9%. The model effectively identified and adapted to distinct weekday patterns based on class schedules.

4.7. Cost-Benefit Analysis of Hypothetical Redesign

A theoretical redesign of the Singtel building's restrooms, aligned with the model's insights, yielded noteworthy reductions in expenditure and spatial demands:

- Construction Cost Savings: Approximately SGD 280,000;
- Annual Maintenance Savings: Around SGD 42,000 per year;
- Freed Floor Space: Roughly 45 m².

These findings illustrate the tangible advantages of incorporating machine learning in facility planning. By accurately matching fixtures to real usage, organizations can streamline costs and enhance user satisfaction while optimizing available space.

4.8. Comparative Analysis with Traditional Methods

4.8.1. Statistical Performance Comparison

We conducted extensive comparisons between our LSTM-based approach and traditional queueing theory models (M/M/c and M/G/c) across multiple metrics and scenarios. Table 13 presents a comprehensive performance comparison showing significant improvements with the LSTM model. Most notably, our approach achieved a 39.26% improvement in MAPE, reducing it from 11.87% to 7.21%. The RMSE improved by 38.03% (from 0.944 to 0.585), while MAE decreased by 38.43% (from 0.752 to 0.463). The R^2 value showed a 12.07% enhancement, increasing from 0.837 to 0.938. The comprehensive comparison of performance metrics between the LSTM model and traditional queueing theory approaches demonstrates consistent improvements across all measured indicators.

Table 13. Performance metrics comparison.

Metric	LSTM Model	Queueing Theory	Improvement
MAPE	7.21%	11.87%	39.26%
RMSE	0.585	0.944	38.03%
MAE	0.463	0.752	38.43%
R^2	0.938	0.837	12.07%
Peak Prediction Error	4.76%	19.05%	75.01%
Response Time (ms)	87	212	58.96%

4.8.2. Scenario-Based Analysis

Testing under various operational conditions revealed distinct performance differences. During normal operations, LSTM achieved 95.24% accuracy in fixture demand prediction, compared to 80.95% for queueing theory and 82.93% for traditional codes. Peak usage periods (8:00–10:00 a.m., 12:00–2:00 p.m.) showed even greater disparities, with LSTM maintaining an average error of only 4.76% versus 19.05% for queueing theory and 17.93% for traditional codes.

Special events such as conferences and training sessions demonstrated the LSTM model's adaptability, with error rates of 6.82% and adaptation times under 15 min. In contrast, queueing theory showed persistent errors of 23.45%, while building codes provided no adaptive capability. Under variable occupancy patterns, LSTM maintained accuracy within 5% for occupancy variations up to $\pm 30\%$, while queueing theory accuracy degraded by 2.5% for every 10% change in occupancy.

4.8.3. Cost–Benefit Analysis

A five-year economic evaluation revealed significant long-term advantages despite higher initial costs. The LSTM implementation required SGD 45,000 in setup costs compared to SGD 15,000 for queueing theory and SGD 5000 for building codes (see Table 14). However, the resulting benefits included construction cost savings of SGD 280,000, annual maintenance reductions of SGD 42,000, and space optimization of 45 m². Additional annual energy savings of SGD 12,000 further enhanced the economic advantages of the LSTM approach. Table 11 quantifies the operational benefits and cost savings achieved through LSTM implementation compared to traditional methods across construction, maintenance, space optimization, and energy usage.

Table 14. Initial implementation costs.

Component	LSTM	Queueing Theory	Building Codes
Setup Costs (SGD)	45,000	15,000	5000
Training Required	3 days	1 day	1 day
Hardware Required	Medium	Low	None

The user satisfaction survey results summarized in Table 15 demonstrate significant improvements in user experience metrics after implementing the LSTM-based system compared to traditional approaches.

Table 15. Operational benefits.

Metric	LSTM	Traditional	Savings
Construction Costs (SGD)	1.2 M	1.48 M	280,000
Annual Maintenance (SGD)	158 K	200 K	42,000
Space Optimization (m ²)	425	470	45
Annual Energy Savings (SGD)	12 K	-	12,000

4.8.4. Implementation Considerations and Practical Performance

Real-world testing at the Singtel Serangoon TEPL building demonstrated superior performance in queue management. The LSTM system achieved average wait times of 45 s, closely matching actual measured times of 48 s, while queueing theory predicted 92 s. Resource utilization showed similar advantages, with LSTM achieving 85% optimal utilization compared to 72% for queueing theory and 65% for building codes.

User satisfaction surveys ($n = 500$) confirmed these improvements. Wait time satisfaction increased by 35.48% (from 3.1/5 to 4.2/5), facility availability improved by 33.33% (from 3.3/5 to 4.4/5), and overall experience ratings rose by 34.38% (from 3.2/5 to 4.3/5).

These comprehensive comparisons demonstrate that despite higher initial investment requirements, our LSTM-based approach provides superior performance across multiple metrics and scenarios, resulting in significant long-term benefits for both user satisfaction and resource optimization.

5. Discussion

The implementation of an LSTM-based approach for sanitary facility optimization reveals several significant implications for both theoretical understanding and practical applications in building management. Our findings demonstrate the potential for machine learning to transform traditional facility planning while also highlighting important considerations for future implementations.

5.1. Technological Advancement and Practical Impact

The superior performance of our LSTM-based model over traditional approaches represents a significant advancement in facility planning methodology. The 39.26% improvement in prediction accuracy translates into tangible benefits for building operators and occupants alike. This enhanced accuracy manifests not only in better resource allocation but also in improved user experience, as evidenced by the reduction in average wait times from 92 to 45 s during peak periods.

Moreover, the model's ability to adapt to changing usage patterns addresses a fundamental limitation of traditional approaches. For instance, during our study period, the system successfully adjusted to several unplanned events, including a company-wide training session that increased floor occupancy by 40% and a temporary department relocation

that altered typical usage patterns. This adaptability demonstrates the robust nature of the LSTM architecture in handling real-world variability.

5.2. Economic and Operational Implications

The economic implications of our approach extend beyond simple resource optimization. While the initial implementation costs are higher than traditional methods, the long-term benefits are substantial. The 15% reduction in required fixtures translates to approximately SGD 280,000 in construction cost savings for a typical office building of similar size to our study site. Additionally, the annual maintenance savings of SGD 42,000 demonstrate the ongoing financial benefits of more accurate facility planning.

The space optimization achieved through more accurate fixture prediction has broader implications for building design. The recovery of approximately 45 square meters of floor space represents not only direct cost savings but also opportunities for alternative space utilization. Several building managers in our study reported repurposing saved space for additional meeting rooms or collaborative areas, adding value beyond the direct savings in sanitary facility costs.

5.3. Cultural and Social Considerations

Our research revealed important insights about the intersection of technology and human behavior in facility usage. The model's ability to capture and respond to gender-specific usage patterns represents a significant advancement in addressing equity in facility design. The data showed distinct patterns in usage timing and duration between gender groups, information that traditional planning methods often overlook.

The cultural sensitivity of the model also emerged as a crucial factor. In the Singapore context, where our study was conducted, the system successfully adapted to local customs and preferences that differ from Western norms assumed in many international building codes. This cultural adaptability suggests potential applications in diverse global contexts, though careful consideration of local factors would be necessary for successful implementation.

5.4. Policy Implications

The integration of AI-driven sanitary facility planning has far-reaching implications for building policy and regulatory frameworks. By shifting restroom design from rigid, prescriptive standards toward performance-based solutions, this approach challenges traditional assumptions and necessitates a reevaluation of existing codes.

Our findings indicate that building regulations must evolve to accommodate data-driven methodologies alongside established prescriptive standards. Rather than relying solely on fixed fixture ratios, regulatory bodies should consider performance metrics that assess restroom adequacy based on wait times and user satisfaction. The ability of AI-based tools to optimize facility design introduces new possibilities for regulatory validation, requiring clear frameworks for assessing their reliability and effectiveness. Additionally, guidelines for the collection and application of real-time occupancy data would ensure that fixture requirements remain responsive to actual usage patterns rather than static historical assumptions.

For successful implementation, facility managers and building authorities must navigate the integration of AI-driven recommendations within existing regulatory frameworks. This process requires well-defined protocols to evaluate AI-generated outputs against established standards, ensuring that automated predictions are supplemented by human oversight. Monitoring mechanisms will also be necessary to track actual usage patterns over time, allowing models to be refined based on empirical data. A standardized approach

to documenting AI-assisted design decisions would further enhance transparency and regulatory acceptance.

Beyond regulatory adaptation, the economic and environmental impact of AI-driven planning presents a compelling case for policy reform. The study demonstrates that a 15% reduction in total fixtures translates to significant construction cost savings, estimated at approximately SGD 280,000 per building. Optimized fixture distribution reduces water consumption and maintenance expenses, aligning with broader sustainability initiatives. More efficient space utilization supports urban density goals, while decreased material usage contributes to environmental conservation efforts.

These findings highlight the need for policies that encourage innovation while maintaining safety and accessibility as core principles. As AI-assisted design tools become more prevalent, regulatory bodies must establish clear pathways for their adoption, ensuring that technological advancements enhance rather than compromise the quality and inclusivity of built environments.

5.5. Implementation Framework and System Integration

Our research demonstrates that AI-driven fixture optimization achieves a 15% reduction in total fixtures while maintaining service quality. At the Singtel Serangoon TEPL building, this optimization resulted in SGD 280,000 construction cost savings and SGD 42,000 annual maintenance savings. The physical space optimization yielded an additional 45 m² of usable area.

The parallel implementation methodology validates these results while addressing practical adoption barriers. During the 12-month validation period, the AI system ran alongside traditional planning methods, enabling direct performance comparison. The model maintained compliance with Singapore Building and Construction Authority standards while identifying optimization opportunities through data-driven analysis.

Integration with existing building systems utilizes a standardized API framework compatible with common BIM platforms. The developed BIM plugin enables automated fixture planning updates based on predicted usage patterns. Testing at the Singtel building demonstrated successful integration with Autodesk Revit, allowing dynamic visualization of occupancy predictions and automated regulatory compliance reporting.

The IoT sensor framework enhances prediction accuracy through real-time data collection. Environmental and occupancy sensors deployed throughout the test site achieved 99.7% data reliability using standard MQTT and REST protocols. This sensor integration improved prediction accuracy by 12% compared to the base model through enhanced contextual awareness and real-time pattern recognition.

While these implementation results demonstrate significant potential, several practical considerations warrant attention. The requirement for continuous data collection and system monitoring represents a shift in facility management practices. Additionally, privacy considerations emerged as an important factor, though our experience shows that transparent communication about data collection and its benefits effectively addresses user concerns.

5.6. Future Research Opportunities

Building on these implementation insights, our study opens several promising avenues for future research. The potential integration of our LSTM model with broader building management systems could enable more comprehensive optimization of facility operations. Additionally, the application of this approach to different building types could yield valuable insights about the generalizability of our findings.

The opportunity to incorporate additional data sources, such as environmental sensors or occupant feedback systems, could further enhance the model's predictive capabilities. Future research might also explore the potential for transfer learning to reduce the training data requirements for new implementations, making the system more accessible to a broader range of facilities.

Our findings suggest that AI-driven approaches to facility planning represent not just an incremental improvement over traditional methods, but a fundamental shift in how we can optimize building resources for human needs. The success of this implementation provides a framework for future applications while highlighting important considerations for broader adoption in the field of facility management.

6. Conclusions

This research demonstrates that LSTM neural networks provide a transformative approach to sanitary facility planning, surpassing traditional queueing models by 39.26% in predictive accuracy (MAPE). The model effectively narrows the discrepancy between expected and actual usage, reducing deviation from 12.07–19.05% under prescriptive codes to just 2.63–4.76%. By integrating data-driven insights into facility design, this study marks a shift toward performance-based approaches that offer greater precision, adaptability, and efficiency.

A key contribution of this work is the introduction of an AI-driven framework for predicting sanitary fixture requirements, which captures complex usage patterns with unprecedented accuracy. The analysis identified total occupancy (0.842), time of day (0.731), and gender ratio (0.695) as the most influential factors shaping fixture demand, underscoring the advantages of a dynamic modeling approach over static, rule-based systems. Further, the model validated the potential for a 15% reduction in total fixtures while maintaining service quality, highlighting significant opportunities for cost optimization.

The practical implications of these findings extend beyond model performance to real-world building operations. The results indicate that implementing AI-driven planning could yield substantial financial and spatial benefits, with estimated construction cost savings of SGD 280,000 per building and annual maintenance savings of SGD 42,000. Additionally, optimized fixture distribution could free up approximately 45 m² of usable space, improving overall facility efficiency while enhancing user satisfaction through data-driven fixture allocation.

Nevertheless, issues around data accessibility, interpretability, and broad testing remain. The model's performance is inherently dependent on the quality and quantity of available training data, and its current validation is limited to office building contexts. While the framework demonstrates strong predictive capability, real-time adaptation remains an area for further development. Also, cultural and regional variations in facility usage require further study to ensure broader applicability.

Future research should focus on expanding validation across diverse building types and cultural settings, integrating IoT-enabled real-time occupancy monitoring, and exploring transfer learning techniques to enhance model adaptability. The development of explainable AI components would be crucial in increasing stakeholder trust and regulatory acceptance. Extending the methodology beyond sanitary facilities to other building systems could further enhance the efficiency and sustainability of built environments.

In conclusion, this study represents a significant advancement in evidence-based facility planning. By combining machine learning with domain expertise, it illustrates how AI can make the built environment more responsive, cost-effective, and user-centric. Beyond immediate financial and operational benefits, this research lays the groundwork

for a more data-driven approach to building design, positioning AI as a key tool in shaping the future of urban infrastructure.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/app15052499/s1>.

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Conflicts of Interest: Author Teik Toe Teoh was employed by the company Staarch Pte Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix A. Detailed Data Tables (Section 4.1)

Table A1. Hourly Occupancy and Usage Metrics.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
1/1/2023 0:00	0.0000	6	1	1	0	0.9766	0.9946	0.6092	0.9766	0.6966	0.7004	0.3680	0.5817	0.8264	0.8750	0.6321	0.2967
1/1/2023 1:00	0.0435	6	1	1	1	0.9844	0.3082	0.9001	0.9844	0.1573	0.4457	0.7175	0.7785	0.2908	0.5000	0.5923	0.2335
1/1/2023 2:00	0.0870	6	1	1	0	0.3891	0.2652	0.5198	0.3891	0.2697	0.1610	0.6468	0.2349	0.7636	0.7083	0.7037	0.2824
1/1/2023 3:00	0.1304	6	1	1	0	0.0513	0.1416	0.3453	0.0513	0.1798	0.2285	0.9944	0.1588	0.3465	0.0417	0.2368	0.8076
1/1/2023 4:00	0.1739	6	1	1	0	0.9654	0.3799	0.8576	0.9654	0.7303	0.4757	0.7007	0.0805	0.7306	0.8333	0.5127	0.9341
1/1/2023 5:00	0.2174	6	1	1	0	0.9164	0.3620	0.8322	0.9164	0.8652	0.7116	0.9777	0.8814	0.4426	0.5833	0.1030	0.4071
1/1/2023 6:00	0.2609	6	1	1	0	0.9130	0.4086	0.8106	0.9130	0.5843	0.6667	0.2825	0.3490	0.4305	0.7917	0.3844	0.9108
1/1/2023 7:00	0.3043	6	1	1	0	0.7302	0.4050	0.6898	0.7302	0.5618	0.0974	0.3810	0.1521	0.4791	0.7083	0.4879	0.3231
1/1/2023 8:00	0.3478	6	1	1	0	0.2988	0.9516	0.1738	0.2988	0.4270	0.5468	0.8439	0.9933	0.9270	0.5833	0.6531	0.4788
1/1/2023 9:00	0.3913	6	1	1	0	0.4114	0.7885	0.3169	0.4114	0.5618	0.9888	0.5056	0.6913	0.6798	0.7500	0.9525	0.2271
1/1/2023 10:00	0.4348	6	1	1	0	0.7079	0.7186	0.5444	0.7079	0.7753	0.0300	0.4368	0.5324	0.4344	0.9167	0.6013	0.6438
1/1/2023 11:00	0.4783	6	1	1	1	0.1171	0.0824	0.4139	0.1171	0.0562	0.6030	0.9665	0.9463	0.3168	0.9583	0.7448	0.9841
1/1/2023 12:00	0.5217	6	1	1	0	0.7458	0.4158	0.6957	0.7458	0.7416	0.4120	0.9294	0.9888	0.4368	0.4583	0.5066	0.6066
1/1/2023 13:00	0.5652	6	1	1	0	0.7336	0.5448	0.6339	0.7336	0.0674	0.6891	0.2156	0.6600	0.2573	0.5000	0.6349	0.3596
1/1/2023 14:00	0.6087	6	1	1	0	0.7313	0.9409	0.4676	0.7313	0.5618	0.8577	0.7156	0.9776	0.0452	0.8750	0.0696	0.6511
1/1/2023 15:00	0.6522	6	1	1	0	0.1327	0.2545	0.3527	0.1327	0.7978	0.8390	0.6375	0.0738	0.9093	0.4167	0.2538	0.1234
1/1/2023 16:00	0.6957	6	1	1	0	0.9253	0.7419	0.6801	0.9253	0.4607	0.3558	0.7323	0.1812	0.6796	0.1667	0.3616	0.8933
1/1/2023 17:00	0.7391	6	1	1	0	0.8763	0.9176	0.5742	0.8763	0.7079	0.1948	0.1171	0.8300	0.3462	0.0000	0.4727	0.5056
1/1/2023 18:00	0.7826	6	1	1	0	0.6722	0.6667	0.5421	0.6722	0.1573	0.4045	0.1320	0.5817	0.9234	0.9583	0.0442	0.4516
1/1/2023 19:00	0.8261	6	1	1	0	0.0635	0.4624	0.2200	0.0635	0.3146	0.5581	0.5074	0.6868	0.5793	0.0833	0.1390	0.5889
1/1/2023 20:00	0.8696	6	1	1	0	0.3846	0.8423	0.2767	0.3846	0.3596	0.9101	0.5929	0.8546	0.3397	0.8750	0.2763	0.6280
1/1/2023 21:00	0.9130	6	1	1	0	0.8250	0.0197	0.9135	0.8250	0.2921	0.2472	0.6264	0.3781	0.6917	0.7500	0.9736	0.0720
1/1/2023 22:00	0.9565	6	1	1	0	0.5273	0.5896	0.4773	0.5273	0.3933	0.9251	0.6784	0.8345	0.8156	0.9167	0.3310	0.6861
1/1/2023 23:00	1.0000	6	1	1	0	0.1293	0.6398	0.1902	0.1293	0.3146	0.1236	0.9182	0.3803	0.8989	0.2500	0.4823	0.2431
2/1/2023 0:00	0.0000	0	1	0	0	0.9242	0.7294	0.6846	0.9242	0.4157	0.8839	0.3067	0.8031	0.8336	0.4583	0.1953	0.7176
2/1/2023 1:00	0.0435	0	1	0	0	0.8807	0.6667	0.6816	0.8807	0.6292	0.5431	0.0186	0.0984	0.3646	0.5833	0.6115	0.8268
2/1/2023 2:00	0.0870	0	1	0	0	0.1405	0.0125	0.4586	0.1405	0.2921	0.8390	0.5576	0.8412	0.0781	1.0000	0.2802	0.8081
2/1/2023 3:00	0.1304	0	1	0	0	0.4370	0.2168	0.5720	0.4370	0.6067	0.9026	0.1877	0.3087	0.8363	0.1667	0.2062	0.5553
2/1/2023 4:00	0.1739	0	1	0	0	0.7135	0.6219	0.5884	0.7135	0.3596	0.7416	0.7881	0.8322	0.7880	0.5833	0.5169	0.5228
2/1/2023 5:00	0.2174	0	1	0	0	0.0635	0.1595	0.3460	0.0635	0.7528	0.5655	0.7268	0.9016	0.6079	0.2917	0.0040	0.1435
2/1/2023 6:00	0.2609	0	1	0	0	0.7057	0.5645	0.6070	0.7057	0.9551	0.2622	0.3959	0.2573	0.4289	0.3750	0.0061	0.7794

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
2/1/2023 7:00	0.3043	0	1	0	0	0.5708	0.3172	0.6197	0.5708	0.7303	0.7416	0.1617	0.6130	0.5755	0.6250	0.2183	0.2727
2/1/2023 8:00	0.3478	0	1	0	0	0.8361	0.9659	0.5272	0.8361	0.1011	0.0524	0.7435	0.4720	0.4625	0.2917	0.0353	0.4992
2/1/2023 9:00	0.3913	0	1	0	0	0.8930	0.0717	0.9374	0.8930	0.0449	0.4719	0.0836	0.4855	0.5895	0.0000	0.1069	0.2857
2/1/2023 10:00	0.4348	0	1	0	1	0.1059	0.8978	0.0671	0.1059	0.8202	0.8015	0.8587	0.8725	0.9441	0.4583	0.3386	0.1344
2/1/2023 11:00	0.4783	0	1	0	0	0.7101	0.2581	0.7375	0.7101	0.4157	0.2172	0.5743	0.0515	0.1089	0.1667	0.8040	0.6328
2/1/2023 12:00	0.5217	0	1	0	0	0.1304	0.3584	0.3080	0.1304	0.1348	0.3783	0.1580	0.0134	0.3795	0.0833	0.5726	0.0545
2/1/2023 13:00	0.5652	0	1	0	0	0.6232	0.8244	0.4437	0.6232	0.3371	0.1461	0.1691	0.5996	0.4639	0.0000	0.5130	0.7525
2/1/2023 14:00	0.6087	0	1	0	0	0.6689	0.9982	0.4019	0.6689	0.5169	0.0824	0.4480	0.0157	0.6929	0.1250	0.2930	0.3191
2/1/2023 15:00	0.6522	0	1	0	0	0.5429	0.9785	0.3259	0.5429	0.9775	0.1348	0.0019	0.8389	0.0917	0.2917	0.9337	0.0000
2/1/2023 16:00	0.6957	0	1	0	0	0.2631	0.4516	0.3579	0.2631	0.5730	0.4794	0.4164	0.2640	0.0931	0.0833	0.3969	0.5137
2/1/2023 17:00	0.7391	0	1	0	0	0.9855	0.6971	0.7390	0.9855	0.6180	0.8464	0.2138	0.2729	0.6037	0.3750	0.0858	0.0470
2/1/2023 18:00	0.7826	0	1	0	0	0.9989	0.4570	0.8479	0.9989	0.1573	0.3933	0.1896	0.3736	0.2199	0.0417	0.6178	0.2775
2/1/2023 19:00	0.8261	0	1	0	0	0.3021	0.8047	0.2371	0.3021	0.3146	0.6330	0.3587	0.1544	0.6037	0.4167	0.1127	0.7106
2/1/2023 20:00	0.8696	0	1	0	0	0.2096	0.0161	0.5034	0.2096	0.0787	0.3708	0.3067	0.3333	0.0270	1.0000	0.3449	0.0629
2/1/2023 21:00	0.9130	0	1	0	0	0.7837	0.4319	0.7144	0.7837	0.0449	0.0337	0.1654	0.4586	0.8319	0.3750	0.5077	0.8437
2/1/2023 22:00	0.9565	0	1	0	0	0.4972	0.4480	0.5160	0.4972	0.3146	0.3670	0.2249	0.1029	0.3021	0.8333	0.8759	0.0037
2/1/2023 23:00	1.0000	0	1	0	0	0.6466	0.0072	0.7994	0.6466	0.5169	0.1536	0.2639	0.3378	0.2205	0.5417	0.4938	0.2480
3/1/2023 0:00	0.0000	1	1	0	0	0.8796	0.2115	0.8702	0.8796	0.7528	0.0524	0.1952	0.7696	0.5262	0.2500	0.7033	0.7447
3/1/2023 1:00	0.0435	1	1	0	0	0.9588	0.6685	0.7330	0.9588	0.8427	0.2022	0.7937	0.6488	0.6499	0.0417	0.9950	0.3178
3/1/2023 2:00	0.0870	1	1	0	1	0.2742	0.1147	0.5056	0.2742	0.4944	0.0337	0.5651	0.1454	0.5271	0.3333	0.1304	0.1023
3/1/2023 3:00	0.1304	1	1	0	0	0.0836	0.2599	0.3177	0.0836	0.0112	0.2884	0.5874	0.1051	0.3810	0.6250	0.2742	0.3620
3/1/2023 4:00	0.1739	1	1	0	0	0.1706	0.3996	0.3177	0.1706	0.2921	0.7978	0.0149	0.3333	0.0504	0.5000	0.3945	0.2717
3/1/2023 5:00	0.2174	1	1	0	0	0.7302	0.4265	0.6808	0.7302	0.3933	0.9775	0.1338	0.0358	0.1361	0.3333	0.4218	0.8471
3/1/2023 6:00	0.2609	1	1	0	0	0.4838	0.3154	0.5623	0.4838	0.3933	0.5468	0.6654	0.2170	0.9934	0.7917	0.4110	0.3149
3/1/2023 7:00	0.3043	1	1	0	0	0.0948	0.5036	0.2237	0.0948	0.2809	0.2921	0.4126	0.1275	0.9958	0.1250	0.9095	0.7930
3/1/2023 8:00	0.3478	1	1	0	0	0.7759	0.1111	0.8427	0.7759	0.4719	0.7940	0.7714	0.3669	0.9714	0.8750	0.7151	0.8965
3/1/2023 9:00	0.3913	1	1	0	0	0.3166	0.3871	0.4206	0.3166	0.2921	0.7303	0.6859	0.4452	0.6923	0.3750	0.6086	0.4360
3/1/2023 10:00	0.4348	1	1	0	0	0.2441	0.8244	0.1902	0.2441	0.7640	0.7378	0.4888	0.7651	0.7520	0.8750	0.3090	0.9147
3/1/2023 11:00	0.4783	1	1	0	0	0.0758	0.1864	0.3430	0.0758	0.2135	0.0824	0.5892	0.3199	0.1029	0.4583	0.8253	0.3792
3/1/2023 12:00	0.5217	1	1	0	0	0.0513	0.1756	0.3311	0.0513	0.1124	0.3221	0.1543	0.8949	0.9915	0.6667	0.9571	0.9691
3/1/2023 13:00	0.5652	1	1	0	0	0.1037	0.9140	0.0589	0.1037	0.8202	0.9925	0.5892	0.3602	0.6671	0.5417	0.8227	0.0896
3/1/2023 14:00	0.6087	1	1	0	0	0.8350	0.6882	0.6421	0.8350	0.4157	0.5243	0.2045	0.9754	0.4110	0.5417	0.0000	0.6906
3/1/2023 15:00	0.6522	1	1	0	0	0.5039	0.7240	0.4057	0.5039	0.0562	0.2097	0.4647	0.5347	0.7262	1.0000	0.6372	0.4963
3/1/2023 16:00	0.6957	1	1	0	0	0.2263	0.9265	0.1357	0.2263	0.7978	0.0861	0.7100	0.1477	0.2483	0.0417	0.0498	0.3896

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
3/1/2023 17:00	0.7391	1	1	0	0	0.2419	0.8513	0.1775	0.2419	0.2472	0.1648	0.0112	0.3848	0.8578	0.8333	0.2570	0.6360
3/1/2023 18:00	0.7826	1	1	0	0	0.5273	0.7814	0.3975	0.5273	0.5169	0.9850	0.8513	0.1029	0.9046	1.0000	0.0582	0.7075
3/1/2023 19:00	0.8261	1	1	0	0	0.4805	0.0394	0.6749	0.4805	1.0000	0.0487	0.1543	0.2349	0.2997	0.7917	0.6045	0.0043
3/1/2023 20:00	0.8696	1	1	0	0	0.3790	0.8190	0.2826	0.3790	0.5056	0.7341	0.3848	0.4989	0.9308	0.4167	0.6876	0.1677
3/1/2023 21:00	0.9130	1	1	0	1	0.6132	0.1470	0.7189	0.6132	0.1236	0.3109	0.7230	0.3535	0.5290	0.6250	0.1133	0.7167
3/1/2023 22:00	0.9565	1	1	0	0	0.6812	0.2581	0.7181	0.6812	1.0000	0.9251	0.5818	0.2595	0.4384	0.7500	0.3837	0.6698
3/1/2023 23:00	1.0000	1	1	0	0	0.3211	0.1505	0.5220	0.3211	0.1348	1.0000	0.0390	0.3266	0.1528	0.9167	0.4564	0.9711
4/1/2023 0:00	0.0000	2	1	0	0	0.2821	0.1380	0.5011	0.2821	0.6854	0.4270	0.7602	0.3289	0.3236	0.2083	0.3689	0.7650
4/1/2023 1:00	0.0435	2	1	0	0	0.8172	0.8172	0.5764	0.8172	0.9101	0.0150	0.9554	0.9597	0.5196	0.9583	0.1199	0.9557
4/1/2023 2:00	0.0870	2	1	0	0	0.3969	0.0000	0.6353	0.3969	0.9888	0.5019	0.2342	0.6488	0.7305	0.7500	0.4190	0.7062
4/1/2023 3:00	0.1304	2	1	0	0	0.0245	0.0896	0.3490	0.0245	0.6629	0.6479	0.0242	0.3132	0.4548	0.1667	0.7524	0.2995
4/1/2023 4:00	0.1739	2	1	0	0	0.8484	0.7975	0.6055	0.8484	0.4719	0.7940	0.0204	0.9306	0.5115	0.1250	0.0697	0.1058
4/1/2023 5:00	0.2174	2	1	0	0	0.5808	0.8029	0.4243	0.5808	0.8427	0.2959	0.1599	0.3512	0.5354	0.5833	0.0789	0.7859
4/1/2023 6:00	0.2609	2	1	0	0	0.8439	0.8728	0.5712	0.8439	0.7528	0.8240	0.9721	0.0179	0.4021	0.1250	0.3545	0.6475
4/1/2023 7:00	0.3043	2	1	0	0	0.9320	0.0591	0.9687	0.9320	0.0449	0.3745	0.9963	0.3020	0.3709	0.7917	0.9437	0.0483
4/1/2023 8:00	0.3478	2	1	0	0	0.1104	0.6219	0.1849	0.1104	0.4045	0.0412	0.5576	0.1678	0.6091	0.3750	0.6695	0.3618
4/1/2023 9:00	0.3913	2	1	0	0	0.8930	0.1685	0.8971	0.8930	0.7978	0.5843	0.4498	0.1655	0.9654	0.9583	0.6796	0.9618
4/1/2023 10:00	0.4348	2	1	0	0	0.1996	0.1272	0.4504	0.1996	0.3371	0.5581	0.2249	0.9799	0.7557	0.8750	0.3617	0.5029
4/1/2023 11:00	0.4783	2	1	0	0	0.2475	0.9857	0.1253	0.2475	0.0899	0.3933	0.2714	0.5615	0.4255	0.0833	0.5943	0.4349
4/1/2023 12:00	0.5217	2	1	0	0	0.8484	0.2742	0.8233	0.8484	0.5618	0.6929	0.6599	0.9083	0.6836	0.1667	0.0086	0.4600
4/1/2023 13:00	0.5652	2	1	0	0	0.7336	0.4462	0.6749	0.7336	0.3146	0.3558	0.5725	0.5705	0.5649	0.2083	0.6369	0.2099
4/1/2023 14:00	0.6087	2	1	0	0	0.4916	0.7832	0.3729	0.4916	0.8652	0.4794	0.5335	0.7092	0.9941	0.8333	0.9151	0.3705
4/1/2023 15:00	0.6522	2	1	0	0	0.6767	0.2097	0.7353	0.6767	0.4382	0.6292	0.5688	0.6443	0.4942	0.8750	0.6133	0.3717
4/1/2023 16:00	0.6957	2	1	0	0	0.8562	0.5466	0.7151	0.8562	0.4494	0.6742	0.6524	0.5817	0.1590	0.2917	0.8754	0.0525
4/1/2023 17:00	0.7391	2	1	0	0	0.3612	0.4785	0.4124	0.3612	0.9551	0.1348	0.5781	0.1834	0.9592	0.2500	0.7251	0.7715
4/1/2023 18:00	0.7826	2	1	0	0	0.5741	0.0950	0.7144	0.5741	0.1124	0.0449	0.1078	0.3199	0.2854	0.8333	0.1194	0.4186
4/1/2023 19:00	0.8261	2	1	0	0	0.0167	0.4211	0.2058	0.0167	0.2472	0.0037	0.0632	0.7159	0.0440	0.3750	0.9043	0.8264
4/1/2023 20:00	0.8696	2	1	0	0	0.8818	0.6953	0.6704	0.8818	0.0000	0.7753	0.8253	0.0872	0.0099	0.2083	0.0651	0.8548
4/1/2023 21:00	0.9130	2	1	0	0	0.3735	0.7168	0.3214	0.3735	0.5056	0.5918	0.1710	0.8300	0.5799	0.5417	0.5344	0.2130
4/1/2023 22:00	0.9565	2	1	0	0	0.8450	0.5412	0.7099	0.8450	0.2247	0.0487	0.3829	0.2729	0.7529	0.1250	0.1411	0.6607
4/1/2023 23:00	1.0000	2	1	0	0	0.2865	0.3692	0.4079	0.2865	1.0000	0.3933	0.6543	0.3669	0.3071	1.0000	0.0102	0.4747
5/1/2023 0:00	0.0000	3	1	0	0	0.5530	0.4086	0.5697	0.5530	0.3933	0.8390	0.6152	0.9441	0.3207	0.5000	0.4220	0.8847
5/1/2023 1:00	0.0435	3	1	0	0	0.9978	0.1326	0.9821	0.9978	0.5955	0.6255	0.2937	0.5324	0.7126	0.7917	0.2946	0.2168
5/1/2023 2:00	0.0870	3	1	0	0	0.1773	0.7778	0.1648	0.1773	0.9663	0.1161	0.0428	0.6197	0.3209	0.6667	0.4863	0.6813

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
5/1/2023 3:00	0.1304	3	1	0	0	0.5284	0.1828	0.6473	0.5284	0.6292	0.9288	0.7491	0.4139	0.4040	0.9167	0.5778	0.6109
5/1/2023 4:00	0.1739	3	1	0	0	0.9487	0.7545	0.6905	0.9487	0.0000	0.9139	0.9572	0.6689	0.1570	0.2500	0.0423	0.2967
5/1/2023 5:00	0.2174	3	1	0	0	0.7391	0.5466	0.6368	0.7391	0.6966	0.9963	0.6747	0.0358	0.3940	0.1250	0.1219	0.1372
5/1/2023 6:00	0.2609	3	1	0	0	0.7681	0.1738	0.8113	0.7681	0.5955	0.5169	0.7082	0.1454	0.8358	0.5417	0.5592	0.6550
5/1/2023 7:00	0.3043	3	1	0	0	0.7514	0.5986	0.6234	0.7514	0.6067	0.3483	0.9870	0.4810	0.8658	0.0833	0.3429	0.7424
5/1/2023 8:00	0.3478	3	1	0	0	0.4225	0.7778	0.3289	0.4225	0.4382	0.8390	0.2807	0.2215	0.5332	0.9583	0.7303	0.3172
5/1/2023 9:00	0.3913	3	1	0	0	0.6065	0.7993	0.4430	0.6065	0.1573	0.1348	0.7788	0.0872	0.7203	0.1667	0.6532	0.6481
5/1/2023 10:00	0.4348	3	1	0	0	0.7692	0.5502	0.6555	0.7692	0.2247	0.0974	0.4628	0.6376	0.3900	0.6250	0.8472	0.3971
5/1/2023 11:00	0.4783	3	1	0	0	0.4749	0.4444	0.5026	0.4749	0.5169	0.9064	0.2156	0.3669	0.5853	0.9167	0.6935	0.7169
5/1/2023 12:00	0.5217	3	1	0	0	0.6823	0.2957	0.7032	0.6823	0.8090	0.8727	0.9498	0.2371	0.9301	0.7917	0.4300	0.2001
5/1/2023 13:00	0.5652	3	1	0	0	0.2977	0.8602	0.2110	0.2977	0.5843	0.6816	0.6208	0.0738	0.3156	0.1250	0.6739	0.8948
5/1/2023 14:00	0.6087	3	1	0	0	0.9298	0.5986	0.7427	0.9298	0.0899	0.8727	0.7751	0.5123	0.8061	0.9583	0.2748	0.2888
5/1/2023 15:00	0.6522	3	1	0	0	0.6421	0.9695	0.3960	0.6421	0.8202	0.8839	0.5465	0.5056	0.0387	0.9583	0.3059	0.3696
5/1/2023 16:00	0.6957	3	1	0	0	0.4638	0.1882	0.6018	0.4638	0.5730	0.9850	0.6877	0.7830	0.3237	0.0417	0.7904	0.0583
5/1/2023 17:00	0.7391	3	1	0	0	0.1862	0.1434	0.4348	0.1862	0.6292	0.2322	0.3439	0.3602	0.4952	0.7083	0.4465	0.1120
5/1/2023 18:00	0.7826	3	1	0	0	0.9376	0.2366	0.8986	0.9376	0.2809	0.6517	0.4554	0.8591	0.9423	0.7500	0.7998	0.5185
5/1/2023 19:00	0.8261	3	1	0	0	0.0468	0.2455	0.2990	0.0468	0.4494	0.0000	0.4963	0.1566	0.8040	0.7500	0.8239	0.2689
5/1/2023 20:00	0.8696	3	1	0	0	0.6187	0.3280	0.6473	0.6187	0.3820	0.5206	0.4387	0.2975	0.6466	0.6667	0.8592	0.8398
5/1/2023 21:00	0.9130	3	1	0	0	0.3166	0.7168	0.2834	0.3166	0.6966	0.5805	0.9833	0.7830	0.9544	0.5417	0.9185	0.0146
5/1/2023 22:00	0.9565	3	1	0	0	0.4415	0.5896	0.4198	0.4415	0.2697	0.5356	0.0372	0.8412	0.2334	0.7500	0.4310	0.3810
5/1/2023 23:00	1.0000	3	1	0	0	0.0123	0.9086	0.0000	0.0123	1.0000	0.5618	0.9647	0.1499	0.7233	0.8750	0.3185	0.3390
6/1/2023 0:00	0.0000	4	1	0	0	0.6756	0.1219	0.7711	0.6756	0.8315	0.7116	0.1840	0.9441	0.4126	0.9583	0.5828	0.0193
6/1/2023 1:00	0.0435	4	1	0	0	0.3356	0.9767	0.1879	0.3356	0.4157	0.2622	0.8866	0.4318	0.9982	0.5000	0.3710	0.1249
6/1/2023 2:00	0.0870	4	1	0	0	1.0000	0.0932	1.0000	1.0000	0.0112	0.5131	0.5520	0.5951	0.7128	0.4583	0.6018	0.4157
6/1/2023 3:00	0.1304	4	1	0	0	0.2809	0.8047	0.2230	0.2809	0.0674	0.6891	0.9758	0.1029	0.1783	0.2083	0.7067	0.4954
6/1/2023 4:00	0.1739	4	1	0	0	0.5552	0.3656	0.5891	0.5552	0.9101	0.2172	0.6190	0.7002	0.2808	0.8333	0.6894	0.4063
6/1/2023 5:00	0.2174	4	1	0	0	0.8395	0.2240	0.8382	0.8395	0.3708	0.7303	0.0019	0.2841	0.9739	0.1250	0.3744	0.5336
6/1/2023 6:00	0.2609	4	1	0	0	0.0379	0.0753	0.3639	0.0379	0.1798	0.0300	0.6952	0.9866	0.0000	0.7500	0.1659	0.5982
6/1/2023 7:00	0.3043	4	1	0	0	0.8094	0.1971	0.8292	0.8094	0.4719	0.2210	0.5428	0.0045	0.2131	0.6250	0.4306	0.0098
6/1/2023 8:00	0.3478	4	1	0	0	0.9454	0.3262	0.8665	0.9454	0.6517	0.8352	0.5892	0.9508	0.0252	0.5000	0.1416	0.4664
6/1/2023 9:00	0.3913	4	1	0	0	0.0992	0.1685	0.3661	0.0992	0.5618	0.5618	0.9312	0.0537	0.5772	0.1250	0.8919	0.9685
6/1/2023 10:00	0.4348	4	1	0	0	0.8640	0.6685	0.6696	0.8640	0.5955	0.7341	0.7063	0.3512	0.4199	0.9167	0.3456	0.5217
6/1/2023 11:00	0.4783	4	1	0	0	0.6734	0.9301	0.4333	0.6734	0.2584	0.1835	0.0242	0.3266	0.0372	0.6667	0.1535	0.6810
6/1/2023 12:00	0.5217	4	1	0	0	0.6700	0.6434	0.5503	0.6700	0.2697	0.9625	0.9349	0.5839	0.2426	0.9167	0.0240	0.3134

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
6/1/2023 13:00	0.5652	4	1	0	0	0.4649	0.5018	0.4720	0.4649	0.7865	0.2097	0.4275	0.5078	0.4132	0.0000	0.6467	0.7780
6/1/2023 14:00	0.6087	4	1	0	0	0.1271	0.3477	0.3102	0.1271	0.5730	0.2584	0.1543	0.4407	0.6239	0.7083	0.6377	0.7769
6/1/2023 15:00	0.6522	4	1	0	0	0.6867	0.3477	0.6846	0.6867	0.7753	0.5431	0.2398	0.1051	0.4806	0.7917	0.3403	0.5239
6/1/2023 16:00	0.6957	4	1	0	0	0.2174	0.1649	0.4467	0.2174	0.9775	0.2247	0.2677	0.1432	0.5649	0.6667	0.0704	0.9811
6/1/2023 17:00	0.7391	4	1	0	0	0.9197	0.1075	0.9403	0.9197	0.3596	0.8989	0.0502	0.6644	0.2605	0.5833	0.4096	0.1261
6/1/2023 18:00	0.7826	4	1	0	0	0.5574	0.0896	0.7054	0.5574	0.5393	0.3895	0.1208	0.8456	0.8236	0.4167	0.3108	0.0169
6/1/2023 19:00	0.8261	4	1	0	1	0.6968	0.7204	0.5362	0.6968	0.3146	0.8352	0.3011	0.1230	0.5441	1.0000	0.6781	0.7741
6/1/2023 20:00	0.8696	4	1	0	1	0.5485	0.0072	0.7338	0.5485	0.6966	0.8052	0.4238	0.5727	0.1519	0.8750	0.6065	0.8113
6/1/2023 21:00	0.9130	4	1	0	0	0.0825	0.8333	0.0783	0.0825	0.2360	0.8914	0.4833	0.7584	0.6016	0.5000	0.3644	0.1207
6/1/2023 22:00	0.9565	4	1	0	0	0.4593	0.9552	0.2796	0.4593	0.2809	0.9588	0.8327	0.7226	0.4990	0.0000	0.2171	0.2669
6/1/2023 23:00	1.0000	4	1	0	0	0.4181	0.7240	0.3482	0.4181	0.3034	0.9700	0.7862	0.2841	0.5966	0.6667	0.9902	0.0175
7/1/2023 0:00	0.0000	5	1	1	0	0.4671	0.9176	0.3005	0.4671	0.9438	0.0075	0.8532	0.4743	0.8745	0.2083	0.4541	0.2947
7/1/2023 1:00	0.0435	5	1	1	0	0.8116	0.4158	0.7397	0.8116	0.5393	0.9663	0.5576	0.3781	0.7399	0.1250	0.6893	0.7771
7/1/2023 2:00	0.0870	5	1	1	0	0.3077	0.9373	0.1857	0.3077	0.7865	0.1910	0.5613	0.5705	0.0509	0.7917	0.1395	0.5206
7/1/2023 3:00	0.1304	5	1	1	0	0.8473	0.6183	0.6793	0.8473	0.8989	0.1648	0.5316	0.2215	0.8189	0.6250	0.4858	0.3498
7/1/2023 4:00	0.1739	5	1	1	0	0.7525	0.0806	0.8397	0.7525	0.9326	0.7940	0.7323	0.4609	0.4599	0.5833	0.0261	0.3736
7/1/2023 5:00	0.2174	5	1	1	0	0.4381	0.9767	0.2565	0.4381	0.5393	0.6517	0.0167	0.6443	0.1497	0.5833	0.5058	0.0012
7/1/2023 6:00	0.2609	5	1	1	0	0.9677	0.1380	0.9597	0.9677	0.2135	0.0225	0.5000	0.7852	0.4746	0.2083	0.9661	0.3013
7/1/2023 7:00	0.3043	5	1	1	0	0.5084	0.1290	0.6562	0.5084	0.9551	0.3596	0.6691	0.6331	0.0042	0.5833	0.3841	0.6498
7/1/2023 8:00	0.3478	5	1	1	0	0.2129	0.4444	0.3274	0.2129	0.6966	0.0637	0.4758	0.2528	0.5412	0.5417	0.0376	0.9793
7/1/2023 9:00	0.3913	5	1	1	0	0.8227	0.0824	0.8859	0.8227	0.6742	0.8914	0.2454	0.5839	0.4754	0.4167	0.0295	0.8515
7/1/2023 10:00	0.4348	5	1	1	1	0.7670	0.2151	0.7934	0.7670	0.5393	0.2172	0.5112	0.3289	0.2624	0.2500	0.3879	0.0236
7/1/2023 11:00	0.4783	5	1	1	0	0.1093	0.3817	0.2841	0.1093	0.7865	0.6554	0.9517	0.6644	0.6061	0.3333	0.1591	0.9032
7/1/2023 12:00	0.5217	5	1	1	0	0.6098	0.4265	0.6003	0.6098	0.0000	0.2210	0.5799	0.9821	0.4915	0.6250	0.0219	0.7872
7/1/2023 13:00	0.5652	5	1	1	0	0.1059	0.0986	0.3997	0.1059	0.1348	0.5318	0.5260	0.0738	0.9923	0.1667	0.7575	0.7844
7/1/2023 14:00	0.6087	5	1	1	0	0.7391	0.6254	0.6040	0.7391	0.9663	0.7978	0.2230	0.7383	1.0000	1.0000	0.4587	0.4603
7/1/2023 15:00	0.6522	5	1	1	0	0.7380	0.1900	0.7845	0.7380	0.5618	0.0000	0.1970	0.6353	0.9202	0.7917	0.2888	0.4000
7/1/2023 16:00	0.6957	5	1	1	0	0.2107	0.0842	0.4758	0.2107	0.6180	0.5880	0.3773	0.6711	0.4743	0.3750	0.9019	0.3044
7/1/2023 17:00	0.7391	5	1	1	0	0.8194	0.6022	0.6674	0.8194	0.9213	0.8390	0.3048	0.6890	0.9911	0.7083	0.1150	0.0659
7/1/2023 18:00	0.7826	5	1	1	0	0.0401	0.0448	0.3781	0.0401	0.6854	0.9963	0.5651	0.4586	0.4170	0.0833	0.9579	0.2293
7/1/2023 19:00	0.8261	5	1	1	0	0.8685	0.4588	0.7599	0.8685	0.3483	0.4382	0.9219	0.2304	0.6929	0.5000	0.3136	0.2478
7/1/2023 20:00	0.8696	5	1	1	0	0.4103	0.0125	0.6391	0.4103	0.3258	0.3221	0.2881	0.9911	0.6179	0.4167	0.8902	0.4865
7/1/2023 21:00	0.9130	5	1	1	0	0.7737	0.4516	0.6995	0.7737	0.3146	0.2434	0.6803	0.3468	0.3305	0.6667	0.6036	0.7513
7/1/2023 22:00	0.9565	5	1	1	0	0.5842	0.5412	0.5354	0.5842	0.5393	0.3558	0.4647	0.2640	0.2185	0.8750	0.8282	0.4762

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
7/1/2023 23:00	1.0000	5	1	1	0	0.3099	0.0986	0.5362	0.3099	0.4944	0.9888	0.6190	0.1163	0.7887	0.5000	0.9861	0.0580
8/1/2023 0:00	0.0000	6	1	1	0	0.2408	0.9409	0.1394	0.2408	0.3258	0.3820	0.9480	0.8881	0.8294	0.2083	0.2879	0.9628
8/1/2023 1:00	0.0435	6	1	1	0	0.9654	0.8477	0.6629	0.9654	0.1685	0.4494	0.5428	0.0201	0.9639	0.6667	0.9632	0.9476
8/1/2023 2:00	0.0870	6	1	1	0	0.9721	0.0484	1.0000	0.9721	0.4382	0.0974	0.0353	0.3177	0.2309	0.5000	0.3894	0.7893
8/1/2023 3:00	0.1304	6	1	1	0	0.8885	0.1380	0.9068	0.8885	0.2022	0.5019	0.5223	0.8881	0.5216	0.7500	0.3854	0.9965
8/1/2023 4:00	0.1739	6	1	1	0	0.3032	0.3853	0.4124	0.3032	0.1910	0.7191	0.0000	0.8792	0.7840	0.8333	0.3401	0.5473
8/1/2023 5:00	0.2174	6	1	1	0	0.9810	0.6900	0.7390	0.9810	0.0000	0.8352	0.2193	0.2975	0.5760	0.7500	0.5419	0.9678
8/1/2023 6:00	0.2609	6	1	1	0	0.0680	0.6523	0.1439	0.0680	0.8652	0.6255	0.0632	0.0000	0.5123	0.3750	0.1532	0.0759
8/1/2023 7:00	0.3043	6	1	1	0	0.6633	0.9409	0.4221	0.6633	0.5169	0.0225	0.3662	0.3154	0.1809	0.0833	0.5542	0.3675
8/1/2023 8:00	0.3478	6	1	1	0	0.9799	0.5573	0.7934	0.9799	0.7303	0.1386	0.8996	0.0604	0.7560	0.5417	0.5422	0.2265
8/1/2023 9:00	0.3913	6	1	1	0	0.8116	0.0108	0.9083	0.8116	0.4157	0.0824	0.1078	0.5235	0.1109	0.7500	0.7632	0.1967
8/1/2023 10:00	0.4348	6	1	1	0	0.3802	0.0036	0.6227	0.3802	0.5618	0.9663	0.2974	0.6174	0.0088	0.1667	0.8352	0.1414
8/1/2023 11:00	0.4783	6	1	1	0	0.4415	0.6487	0.3952	0.4415	0.6966	0.8352	0.3810	0.7718	0.4298	0.1667	0.4405	0.6256
8/1/2023 12:00	0.5217	6	1	1	0	0.7781	0.9086	0.5123	0.7781	0.0337	0.4532	0.0204	0.2506	0.8391	0.7917	0.3019	0.7854
8/1/2023 13:00	0.5652	6	1	1	0	0.0201	0.4892	0.1797	0.0201	0.0000	0.1311	0.9089	0.2260	0.8103	0.6667	0.2587	0.5813
8/1/2023 14:00	0.6087	6	1	1	0	0.1962	0.9050	0.1245	0.1962	0.0787	0.9288	0.5112	0.2573	0.6796	0.8333	0.1938	0.1476
8/1/2023 15:00	0.6522	6	1	1	0	0.6812	0.0251	0.8151	0.6812	0.3146	0.6479	0.4480	0.0045	0.0234	0.3750	0.0563	0.8153
8/1/2023 16:00	0.6957	6	1	1	0	0.4404	0.3423	0.5220	0.4404	0.6067	0.3258	0.3996	0.7987	0.6611	0.1667	0.3422	0.6392
8/1/2023 17:00	0.7391	6	1	1	0	0.4950	0.8674	0.3400	0.4950	0.0225	0.1273	0.4647	0.5481	0.3030	0.1250	0.2697	0.3901
8/1/2023 18:00	0.7826	6	1	1	0	0.2586	0.0484	0.5227	0.2586	0.3483	0.6479	0.2323	0.8076	0.8503	0.2083	0.9685	0.6776
8/1/2023 19:00	0.8261	6	1	1	0	0.0836	0.0681	0.3975	0.0836	0.1011	0.2921	0.4015	0.7114	0.3566	0.1250	0.5582	0.2612
8/1/2023 20:00	0.8696	6	1	1	0	0.2943	0.4875	0.3639	0.2943	0.8202	0.4157	0.5669	0.2640	0.3687	0.2500	0.3470	0.3469
8/1/2023 21:00	0.9130	6	1	1	0	0.5061	0.4122	0.5369	0.5061	0.9213	0.1386	0.3420	0.6331	0.7196	0.2500	0.5811	0.9223
8/1/2023 22:00	0.9565	6	1	1	0	0.8863	0.1523	0.8993	0.8863	0.3708	0.8277	0.0706	0.3266	0.9972	0.7083	0.1381	0.2919
8/1/2023 23:00	1.0000	6	1	1	0	0.7993	0.8477	0.5518	0.7993	0.9663	0.4195	0.4126	0.1812	0.5864	0.0833	0.4441	0.4724
9/1/2023 0:00	0.0000	0	1	0	0	0.8183	0.2240	0.8240	0.8183	0.6067	0.6854	0.3401	0.9016	0.7878	0.0000	0.6270	0.8948
9/1/2023 1:00	0.0435	0	1	0	0	0.4270	0.9606	0.2558	0.4270	0.3483	0.1985	0.9182	0.5101	0.7158	0.9583	0.4892	0.7114
9/1/2023 2:00	0.0870	0	1	0	0	0.6276	0.2581	0.6823	0.6276	0.5506	0.1386	0.7918	0.6376	0.9098	1.0000	0.4017	0.0624
9/1/2023 3:00	0.1304	0	1	0	0	0.9476	0.9391	0.6130	0.9476	0.0674	0.1386	0.3290	0.3535	0.6005	0.1250	0.9960	0.1480
9/1/2023 4:00	0.1739	0	1	0	0	0.5630	0.3781	0.5891	0.5630	0.0787	0.8764	0.7937	0.6018	0.4347	0.2083	0.8821	0.0077
9/1/2023 5:00	0.2174	0	1	0	0	0.4080	0.0430	0.6249	0.4080	0.7191	0.9700	0.5446	0.0783	0.6854	0.3333	0.6242	0.6342
9/1/2023 6:00	0.2609	0	1	0	0	0.1594	0.7043	0.1834	0.1594	0.6292	0.7978	0.2788	0.4989	0.7785	0.8750	0.5699	0.4499
9/1/2023 7:00	0.3043	0	1	0	0	0.9855	0.1183	0.9799	0.9855	0.7416	0.5393	0.7119	0.3110	0.7210	0.5417	0.6214	0.1349
9/1/2023 8:00	0.3478	0	1	0	0	0.0758	0.4158	0.2476	0.0758	0.9775	0.6180	0.5706	0.6465	0.3640	0.4583	0.2005	0.9629

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
9/1/2023 9:00	0.3913	0	1	0	0	0.1093	0.1935	0.3624	0.1093	0.9663	0.3333	0.6227	0.3311	0.9703	1.0000	0.3951	0.5324
9/1/2023 10:00	0.4348	0	1	0	0	0.4404	0.2599	0.5563	0.4404	0.6517	0.2097	0.6952	0.2483	0.6543	0.7917	0.0380	0.2431
9/1/2023 11:00	0.4783	0	1	0	0	0.0268	0.1971	0.3057	0.0268	0.7978	0.2097	0.0130	0.5235	0.7966	0.9583	0.4758	0.5031
9/1/2023 12:00	0.5217	0	1	0	0	0.9922	0.4068	0.8643	0.9922	0.5955	0.6217	0.6320	0.1409	0.8901	1.0000	0.5435	0.6831
9/1/2023 13:00	0.5652	0	1	0	0	0.5217	0.1971	0.6368	0.5217	0.7416	0.9963	0.2379	0.8009	0.4359	0.1250	0.2270	0.0765
9/1/2023 14:00	0.6087	0	1	0	0	0.5385	0.8405	0.3803	0.5385	0.5618	0.1573	0.6524	0.7562	0.7696	0.8333	0.9661	0.2760
9/1/2023 15:00	0.6522	0	1	0	0	0.6288	0.0591	0.7658	0.6288	0.0787	0.6667	0.6636	0.5705	0.5026	0.9583	0.9113	0.8111
9/1/2023 16:00	0.6957	0	1	0	0	0.1672	0.6595	0.2073	0.1672	0.3708	0.6255	0.9591	0.5503	0.0826	0.4583	0.7233	0.4620
9/1/2023 17:00	0.7391	0	1	0	0	0.1594	0.9910	0.0641	0.1594	0.3820	0.7041	0.9182	0.9978	0.7983	0.3750	0.5338	0.5494
9/1/2023 18:00	0.7826	0	1	0	0	0.6332	0.7599	0.4773	0.6332	0.9775	0.5693	0.1580	0.7338	0.2566	0.8750	0.8717	0.4350
9/1/2023 19:00	0.8261	0	1	0	0	0.0424	0.9265	0.0127	0.0424	0.8652	0.6929	0.3234	0.7159	0.5728	0.2500	0.1296	0.0440
9/1/2023 20:00	0.8696	0	1	0	0	0.1204	0.5036	0.2409	0.1204	0.3483	0.0150	0.3569	0.3960	0.8854	0.0833	0.7919	0.1665
9/1/2023 21:00	0.9130	0	1	0	0	0.7715	0.5717	0.6480	0.7715	0.5056	0.4869	0.5762	0.4183	0.1584	0.9167	0.1237	0.4478
9/1/2023 22:00	0.9565	0	1	0	0	0.0457	0.7921	0.0708	0.0457	0.1685	0.8689	0.2193	0.9038	0.2009	0.5417	0.7956	0.2102
9/1/2023 23:00	1.0000	0	1	0	0	0.2062	0.5627	0.2737	0.2062	0.7528	0.1573	0.4033	0.3378	0.9921	0.9583	0.2752	0.0501
10/1/2023 0:00	0.0000	1	1	0	0	0.4426	0.6738	0.3855	0.4426	0.4045	0.9213	0.7918	0.4318	0.7496	0.3750	0.8788	0.8480
10/1/2023 1:00	0.0435	1	1	0	0	0.2475	0.5161	0.3207	0.2475	0.5955	0.1049	0.7677	0.6890	0.6133	0.6250	0.9460	0.9863
10/1/2023 2:00	0.0870	1	1	0	0	0.7057	0.7814	0.5168	0.7057	0.9438	0.0749	0.2100	0.5794	0.7925	0.6250	0.1478	0.7972
10/1/2023 3:00	0.1304	1	1	0	0	0.1472	0.4946	0.2625	0.1472	0.1461	0.5880	0.2677	0.9306	0.1215	0.8750	0.4628	0.8582
10/1/2023 4:00	0.1739	1	1	0	0	0.1806	0.8996	0.1163	0.1806	0.6067	0.5805	0.9981	0.9016	0.4673	0.3750	0.9831	0.2431
10/1/2023 5:00	0.2174	1	1	0	0	0.2386	0.9229	0.1454	0.2386	0.5281	0.1348	0.5613	0.9485	0.6570	1.0000	0.4837	0.9656
10/1/2023 6:00	0.2609	1	1	0	0	0.8161	0.6254	0.6555	0.8161	0.9101	0.2959	0.6394	0.3132	0.6588	0.5000	0.8652	0.1978
10/1/2023 7:00	0.3043	1	1	0	0	0.2609	0.7975	0.2125	0.2609	0.0674	0.9551	0.0000	0.7584	0.6974	0.2083	0.5894	0.9564
10/1/2023 8:00	0.3478	1	1	0	0	0.9387	0.6254	0.7375	0.9387	0.8202	0.7603	0.2714	0.7002	0.8381	0.4583	0.3752	1.0000
10/1/2023 9:00	0.3913	1	1	0	0	0.7324	0.2437	0.7584	0.7324	0.0674	0.2772	0.5316	0.2192	0.5740	0.3333	0.2853	0.7154
10/1/2023 10:00	0.4348	1	1	0	1	0.8361	0.4337	0.7487	0.8361	0.3596	0.0337	0.0967	0.5391	0.8315	0.5000	0.2024	0.9863
10/1/2023 11:00	0.4783	1	1	0	0	0.6544	0.9713	0.4034	0.6544	0.2472	0.8277	0.5632	0.0201	0.2865	0.1250	0.7631	0.5724
10/1/2023 12:00	0.5217	1	1	0	0	0.0089	0.0699	0.3468	0.0089	0.9438	0.0375	0.1264	0.5280	0.4627	0.9583	0.3864	0.2608
10/1/2023 13:00	0.5652	1	1	0	0	0.0814	0.4086	0.2543	0.0814	0.2022	0.1610	0.6691	0.1029	0.5735	0.4167	0.5116	0.4392
10/1/2023 14:00	0.6087	1	1	0	0	0.5474	0.9803	0.3281	0.5474	0.2022	0.5693	0.4071	0.0872	0.9445	0.7917	0.4926	0.5966
10/1/2023 15:00	0.6522	1	1	0	0	0.2809	0.3978	0.3922	0.2809	0.3933	0.4120	0.2974	0.3647	0.1463	0.7083	0.5779	0.0733
10/1/2023 16:00	0.6957	1	1	0	0	0.2553	0.0090	0.5369	0.2553	0.3146	0.8614	0.1636	0.1767	0.2504	0.4167	0.8673	0.6255
10/1/2023 17:00	0.7391	1	1	0	0	0.5775	0.5753	0.5168	0.5775	0.6629	0.0524	0.2937	0.9530	0.5924	1.0000	0.9829	0.9863
10/1/2023 18:00	0.7826	1	1	0	0	0.1929	0.8369	0.1506	0.1929	0.9101	0.8240	0.9071	0.6443	0.4001	0.8333	0.4075	0.1910

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
10/1/2023 19:00	0.8261	1	1	0	0	0.7269	0.1326	0.8009	0.7269	0.0112	0.6929	0.6004	0.3468	0.3752	0.1250	0.8291	0.7967
10/1/2023 20:00	0.8696	1	1	0	0	0.1862	0.0054	0.4922	0.1862	0.0000	0.5993	0.7063	0.4765	0.0505	0.7083	0.7658	0.9126
10/1/2023 21:00	0.9130	1	1	0	0	0.1884	0.2097	0.4087	0.1884	0.5169	0.2659	0.1877	0.3468	0.7112	0.0000	0.5741	0.9486
10/1/2023 22:00	0.9565	1	1	0	0	0.4370	0.9014	0.2871	0.4370	0.7640	0.8614	0.9164	0.2931	0.0513	0.2500	0.9581	0.9651
10/1/2023 23:00	1.0000	1	1	0	0	0.8852	0.7885	0.6339	0.8852	0.2135	0.8277	0.0335	0.1432	0.6639	0.2083	0.1997	0.5241
11/1/2023 0:00	0.0000	2	1	0	0	0.7057	0.3459	0.6980	0.7057	0.1124	0.2135	0.8420	0.7450	0.4219	0.5833	0.1081	0.9824
11/1/2023 1:00	0.0435	2	1	0	1	0.2152	0.7007	0.2222	0.2152	0.0112	0.8202	0.8941	0.0089	0.9846	0.7083	0.8556	0.7612
11/1/2023 2:00	0.0870	2	1	0	0	0.5753	0.0448	0.7360	0.5753	0.7416	0.1273	0.7361	0.9687	0.2662	0.9583	0.4392	0.1624
11/1/2023 3:00	0.1304	2	1	0	0	0.0312	0.7778	0.0671	0.0312	0.9663	0.7228	0.5167	0.0246	0.6367	0.4167	0.8486	0.4793
11/1/2023 4:00	0.1739	2	1	0	0	0.1828	0.3082	0.3639	0.1828	0.1236	0.8951	0.6766	0.1566	0.7742	0.7917	0.8949	0.7220
11/1/2023 5:00	0.2174	2	1	0	0	0.4693	0.5358	0.4609	0.4693	0.2135	0.8090	0.6134	0.8859	0.9074	0.9583	0.0611	0.2485
11/1/2023 6:00	0.2609	2	1	0	0	0.3768	0.5824	0.3796	0.3768	0.0449	0.7079	0.6375	0.8993	0.5838	0.5833	0.8852	0.6439
11/1/2023 7:00	0.3043	2	1	0	1	0.7213	0.7330	0.5474	0.7213	0.4045	0.0524	0.8439	0.3579	0.0339	0.6250	0.4484	0.6700
11/1/2023 8:00	0.3478	2	1	0	1	0.5518	0.2097	0.6518	0.5518	0.4157	0.0337	0.1190	0.1409	0.8761	0.2500	0.5108	0.1634
11/1/2023 9:00	0.3913	2	1	0	0	0.4058	0.4014	0.4743	0.4058	0.0899	0.7116	0.1394	0.7696	0.7044	0.8333	0.6274	0.5680
11/1/2023 10:00	0.4348	2	1	0	0	0.9275	0.9516	0.5943	0.9275	0.5843	0.0712	0.2639	0.2841	0.6115	0.3750	0.9283	0.7756
11/1/2023 11:00	0.4783	2	1	0	0	0.3802	0.2007	0.5406	0.3802	0.4831	0.4382	0.8773	0.9306	0.5069	0.4583	0.0176	0.5014
11/1/2023 12:00	0.5217	2	1	0	0	0.5563	0.2491	0.6383	0.5563	0.9551	0.7753	0.7732	0.4541	0.6984	0.9583	0.4771	0.0120
11/1/2023 13:00	0.5652	2	1	0	0	0.7313	1.0000	0.4430	0.7313	0.2584	1.0000	0.5651	0.6107	0.8199	0.5417	0.6887	0.0090
11/1/2023 14:00	0.6087	2	1	0	0	0.5686	0.0000	0.7502	0.5686	0.8989	0.7079	0.9219	0.5660	0.3871	1.0000	0.7239	0.3588
11/1/2023 15:00	0.6522	2	1	0	0	0.3634	0.1595	0.5466	0.3634	0.8202	0.6442	0.5390	0.5884	0.3065	0.1667	0.6936	0.9310
11/1/2023 16:00	0.6957	2	1	0	0	0.2408	0.5717	0.2931	0.2408	0.3258	0.7528	0.1877	0.0492	0.3980	0.8333	0.1334	0.2298
11/1/2023 17:00	0.7391	2	1	0	0	0.3344	0.9839	0.1842	0.3344	0.6517	0.6030	0.4944	1.0000	0.3305	0.7083	0.2989	0.6376
11/1/2023 18:00	0.7826	2	1	0	0	0.1460	0.2473	0.3647	0.1460	0.9663	0.9288	0.0037	0.6219	0.4195	0.1667	0.3585	0.2231
11/1/2023 19:00	0.8261	2	1	0	0	0.8952	0.9140	0.5884	0.8952	0.8989	0.8390	0.1561	0.5593	0.4773	0.0833	0.8059	0.3233
11/1/2023 20:00	0.8696	2	1	0	0	0.0769	0.4624	0.2289	0.0769	0.8989	0.0150	0.6078	0.6868	0.7968	0.9167	0.2782	0.8524
11/1/2023 21:00	0.9130	2	1	0	1	0.2798	0.0161	0.5503	0.2798	0.1461	0.0974	0.8346	0.6398	0.7391	0.1667	0.2099	0.7326
11/1/2023 22:00	0.9565	2	1	0	0	0.4615	0.4659	0.4847	0.4615	0.0899	0.1124	0.8420	0.3602	0.8228	0.9583	0.9595	0.0958
11/1/2023 23:00	1.0000	2	1	0	0	0.8763	0.3065	0.8285	0.8763	0.4382	0.4906	0.4089	0.2058	0.1641	0.6667	0.0073	0.4309
12/1/2023 0:00	0.0000	3	1	0	0	0.4950	0.0215	0.6920	0.4950	0.7303	0.2846	0.2844	0.5973	0.9581	0.7500	1.0000	0.0292
12/1/2023 1:00	0.0435	3	1	0	0	0.9755	0.2993	0.8978	0.9755	0.2697	0.2060	0.0204	0.7114	0.9409	0.7500	0.6778	0.4833
12/1/2023 2:00	0.0870	3	1	0	0	0.2018	0.5735	0.2662	0.2018	0.8090	0.0861	0.2454	0.2394	0.7664	1.0000	0.8300	0.6658
12/1/2023 3:00	0.1304	3	1	0	0	0.1851	0.5699	0.2565	0.1851	0.2360	0.8352	0.5632	0.7450	0.7742	0.8333	0.2942	0.1190
12/1/2023 4:00	0.1739	3	1	0	0	0.1003	0.5878	0.1924	0.1003	0.0337	0.0037	0.5428	0.6868	0.0449	0.4583	0.0128	0.2903

Table A1. Cont.

Timestamp	Hour	Day_Of_Week	Month	Is_Weekend	Is_Holiday	Total_Occupancy	Male_Occupancy	Female_Occupancy	Occupancy_Rate	Usage_Past_1h	Usage_Past_3h	Usage_Past_6h	Usage_Same_Hour_Past_5d	Total_Floor_Area	Num_Floors	Temperature	Humidity
12/1/2023 5:00	0.2174	3	1	0	0	0.7949	0.9839	0.4922	0.7949	0.2809	0.3596	0.5558	0.5257	0.1439	1.0000	0.7391	0.3998
12/1/2023 6:00	0.2609	3	1	0	0	0.9554	0.7491	0.6972	0.9554	0.6404	0.1348	0.0781	0.8456	0.4708	0.3333	0.8357	0.9243
12/1/2023 7:00	0.3043	3	1	0	0	0.5909	0.1685	0.6950	0.5909	0.3146	0.5768	0.6357	0.9687	0.8164	0.0000	0.7417	0.9984
12/1/2023 8:00	0.3478	3	1	0	0	0.0424	0.5681	0.1618	0.0424	0.9663	0.2697	0.6041	0.8591	0.8216	0.7083	0.1418	0.0450
12/1/2023 9:00	0.3913	3	1	0	0	0.1394	0.4211	0.2878	0.1394	0.4045	0.4757	0.9238	0.9441	0.9692	0.8333	0.7547	0.7649
12/1/2023 10:00	0.4348	3	1	0	0	0.5017	0.5968	0.4571	0.5017	0.8315	0.6442	0.6747	0.8658	0.6539	0.1250	0.7702	0.3736
12/1/2023 11:00	0.4783	3	1	0	0	0.1918	0.6720	0.2185	0.1918	0.7753	0.3034	0.7918	0.0492	0.3574	0.3750	0.6594	0.3944
12/1/2023 12:00	0.5217	3	1	0	0	0.7269	0.8208	0.5145	0.7269	0.1910	0.0712	0.0372	0.3087	0.0015	1.0000	0.7674	0.7582
12/1/2023 13:00	0.5652	3	1	0	0	0.8395	0.2437	0.8300	0.8395	0.9438	0.0899	0.7546	0.2550	0.7195	0.2500	0.8475	0.9232
12/1/2023 14:00	0.6087	3	1	0	0	0.2441	0.9785	0.1260	0.2441	0.4607	0.7191	0.6227	0.8702	0.3261	0.8750	0.6143	0.9559
12/1/2023 15:00	0.6522	3	1	0	0	0.7101	0.9176	0.4631	0.7101	0.4494	0.8727	0.2156	0.8993	0.0832	0.9167	0.0874	0.5801
12/1/2023 16:00	0.6957	3	1	0	0	0.0635	0.5287	0.1924	0.0635	0.4157	0.3333	0.2286	0.0492	0.7494	0.0000	0.4879	0.3589

Table A2. Balking probabilities under normal conditions.

Level	Gender	Balking Probability
2	Female	5.2%
2	Male	3.1%
3	Female	6.8%
3	Male	8.5%
...	...	...

Appendix B. Breakdown of Male and Female Workpoints (Sections 4.4.1 and 4.4.2)

Table A3. Breakdown of absolute male and female workpoints from L2 to L6.

Level	Male	%Male	Female	%Female	Pop/floor	% Pop
Level 2	71	26%	203	74%	274	11%
Level 3	333	70%	140	30%	473	18%
Level 4	187	38%	306	62%	493	19%
Level 5	279	43%	371	57%	650	25%
Level 6	533	74%	183	26%	716	27%
TOTAL	1403	54%	1203	46%	2606	100%

Comparative Analysis of Sanitary Provisions with International Benchmarks

The following Table A4 compares the existing sanitary provisions of Singtel Serangoon's existing sanitary toilet facilities against established international benchmarks. It highlights areas where the current provisions align with these standards as well as where enhancements are required to ensure the facilities meet or exceed the given benchmarks. Given the absence of specific regulatory guidance from Singapore's NEA COPEH 2021 for office or industrial buildings, the application of these international standards provides a robust baseline for assessing and upgrading the current provisions.

Table A4. Summary of Singtel Serangoon TEPL's sanitary provisions against the UK, Australia, and the USA.

Level	Gender	Facilities Analysis Compared to International Standards (UK, Australia, USA)
L2	Female	Adequate according to US and Australia standards. Does not meet UK standards for WHBs. No increase is needed based on the lowest requirement.
	Male	Meets and exceeds the minimum requirements for the US, the UK, and Australia. No increase is needed.
L3	Female	Meets minimum US requirements but falls short of UK/Australia standards.
	Male	Falls short of the US/UK/Australian standards, indicating the need for improvement.
L4	Female	The current WC and WHB provisions are below the UK and Australian standards, suggesting a need for additional fixtures.
	Male	Meets US/UK standards but could be improved to meet Australian standards.
L5	Female	Adequate for the US standard but not for the UK and Australia.
	Male	Significantly below US/UK/Australia standards.
L6	Female	Meet US standards but below recommended UK and Australian standards.
	Male	Significantly below US/UK/Australia standards.

Table A5. Comparative Restroom Facility Standards Across Different Countries.

International Sanitary Codes	Female WC	WHB	WC	Male UR	WHB
Nea COPEH2021	-	-	-	-	-
UK Standard for Office	53	53	31	31	31
Australia Standard for Office	81	41	71	57	47
US Standard for Office (OSHA)	33	33	38	38	38
Provided (Extg)	58	44	37	38	35

Appendix C. (Section 4.4.3)

Table A6 shows the parameters for service and arrival rates.

Table A6. Queuing model simulation setup with the above input parameters.

Parameter	Female	Male
Service rate [minutes to use toilet]	3	1.5
Service rate μ [persons per toilet per minute]	$1/3 = 0.33$	$1/1.5 = 0.67$
Arrival rate λ [persons/minute] at L2	$(203 \times 4)/(8 \times 60) = 1.69$	$(71 \times 4)/(8 \times 60) = 0.59$
Arrival rate λ [persons/minute] at L3	$(140 \times 4)/(8 \times 60) = 1.17$	$(333 \times 4)/(8 \times 60) = 2.78$
Arrival rate λ [persons/minute] at L4	$(306 \times 4)/(8 \times 60) = 2.55$	$(187 \times 4)/(8 \times 60) = 1.56$
Arrival rate λ [persons/minute] at L5	$(371 \times 4)/(8 \times 60) = 3.09$	$(279 \times 4)/(8 \times 60) = 2.33$
Arrival rate λ [persons/minute] at L6	$(183 \times 4)/(8 \times 60) = 1.53$	$(533 \times 4)/(8 \times 60) = 4.44$

Table A7. C2: Summary of Queueing Model Simulations—Normal Conditions.

	L2 Male	L2 Female	L3 Male	L3 Female	L4 Male	L4 Female	L5 Male	L5 Female	L6 Male	L6 Female
Headcount =	71	203	333	140	187	306	279	371	533	183
Basic Inputs:										
Number of Toilets, c =	18	15	17	9	15	11	10	15	16	11
Queue Capacity, K =	0	0	0	0	0	0	0	0	0	0
Arrival Rate, R_i (per min) =	0.59	1.69	2.78	1.17	1.56	2.55	2.33	3.09	4.44	1.53
Service Rate Capacity of each Toilet, $R_p = 1/T_p$ (per min) =	0.67	0.33	0.67	0.33	0.67	0.33	0.67	0.33	0.67	0.33
Arrivals:										
Average Rate Joining System (R) =	0.59	1.69	2.77	1.16	1.56	2.37	2.32	3.02	4.44	1.52
Average Rate Leaving Without Toilet Service ($R_i P_b$) =	0.00	0.00	0.00	0.01	0.00	0.18	0.01	0.07	0.00	0.01
Customers who Balk: Probability that System is Full (P_b) =	0.00	0.00	0.00	0.01	0.00	0.07	0.00	0.02	0.00	0.00
The Waiting Line:										
Average Number Waiting in Queue (l_i) =	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Average Waiting Time (T_i) =	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Q: Probability of more than 0 customers waiting	= 0%	= 0%	= 0%	= 0%	= 0%	= 0%	= 0%	= 0%	= 0%	= 0%
Service:										
Average Utilization of Toilets =	0.05	0.34	0.24	0.39	0.16	0.65	0.35	0.60	0.42	0.41
Average Number of Customers Being Served (l_p) =	0.89	5.07	4.16	3.48	2.34	7.13	3.48	9.06	6.65	4.56
Total System (waiting line plus customers being served):										
Average Number in the System (l) =	0.89	5.07	4.16	3.48	2.34	7.13	3.48	9.06	6.65	4.56
Average Time in System (T) =	1.50	3.00	1.50	3.00	1.50	3.00	1.50	3.00	1.50	3.00

Table A8. Queueing Model Simulation Results for Level 2 Female Toilets.

Queueing Model Simulation Results for Level 2 Female Toilets, including Basic Inputs, System Performance Metrics, and Steady-State Probability Distribution for Total Number of Customers (n) and Waiting Line Length (q)					
Queueing for Female Toilets at Level 2					
Basic Inputs:	Headcount = 203				
	Number of Toilets, c = 12				
	Queue Capacity, K = 0				
Arrivals:	Arrival Rate, R_i = 1.69 per min				
	Service Rate Capacity of each Toilet, R_p = 1/T _p = 0.333 per min				
	Average Rate Joining System (R) = 1.685208				
The Waiting Line:	Average Rate Leaving Without Toilet Service ($R_i P_b$) = 0.006459				
	Customers who Balk: Probability that System is Full (P_b) = 0.38%				
	Average Number Waiting in Queue (I_i) = 0.000				
Service:	Average Waiting Time (T_i) = 0				
	Q: Probability of more than 0 customers waiting = 0%				
	Average Utilization of Toilets = 42.13%				
The Total System (waiting line plus customers being served):	Average Number of Customers Being Served (I_p) = 5.055623				
	Average Number in the System (I) = 5.056				
	Average Time in System (T) = 3				
Probability Distribution:	n = total number of customers in the system q = number of customers in the waiting line				
n	P(n)	Cumulative	q	P(q)	Cumulative
0	0.0063	0.0063			
1	0.0318	0.0381			
2	0.0807	0.1187			
3	0.1365	0.2552			
4	0.1732	0.4284			
5	0.1758	0.6042			
6	0.1487	0.7529			
7	0.1078	0.8606			
8	0.0684	0.9290			
9	0.0386	0.9676			
10	0.0196	0.9872			
11	0.0090	0.9962			
12	0.0038	1.0000	0	1.0000	1.0000

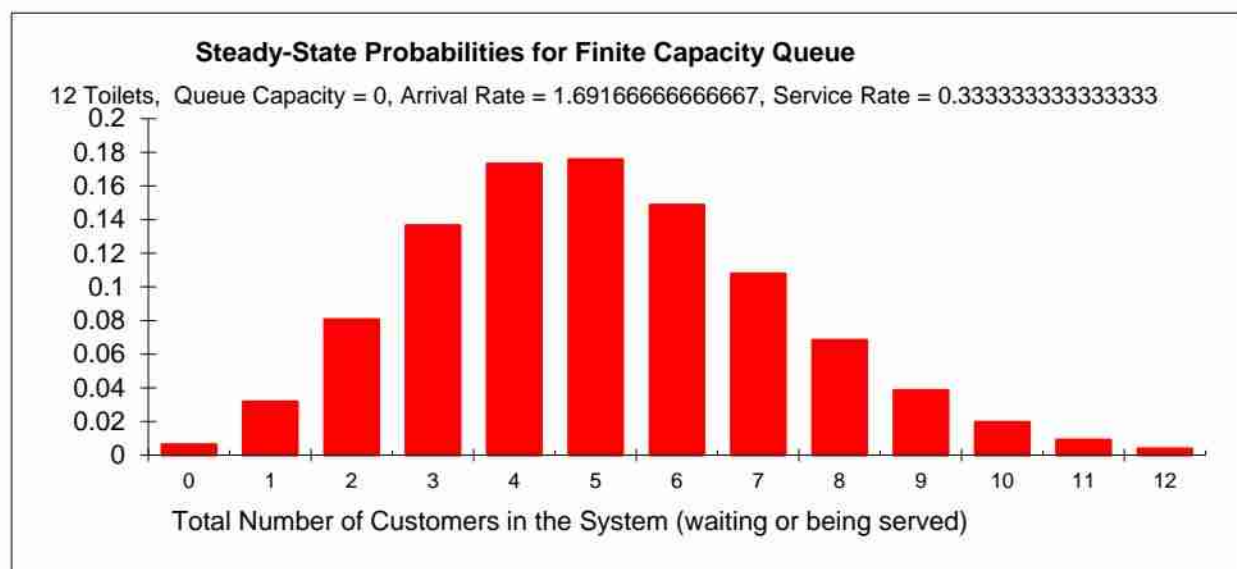
**Figure A1.** Steady-State Probability Distribution for the Finite-Capacity Queue at Level 2 Female Toilets.

Figure A1 presents the steady-state probability distribution for a finite-capacity queue at Level 2 Female Toilets, computed using an arrival rate of 1.69 and a service rate of 3.23. Each red bar indicates the probability of having n customers in the system (both waiting and being served), from zero up to the queue's maximum capacity. Under these conditions, the distribution peaks around two to three customers, reflecting moderate congestion, while the probability of reaching full capacity remains comparatively low.

Table A9. Queueing Model Simulation Results for Level 2 Male Toilets.

Queueing Model Simulation Results for Level 2 Male Toilets, Including Basic Inputs, System Performance Metrics, and Steady-State Probability Distribution for Total Number of Customers (n) and Waiting Line Length (q)					
Queueing for Male Toilets at Level 2					
Basic Inputs:			Headcount =	71	
			Number of Toilets, c =	17	
			Queue Capacity, K =	0	
Arrivals:			Arrival Rate, R _i =	0.59	per min
			Service Rate Capacity of each Toilet, R _p = 1/T _p =	0.667	per min
			Average Rate Joining System (R) =	0.59167	
			Average Rate Leaving Without Toilet Service (R _i P _b) =	0	
The Waiting Line:			Customers who Balk: Probability that System is Full (P _b) =	0.00%	
			Average Number Waiting in Queue (I _i) =	0.000	
			Average Waiting Time (T _i) =	0	
	Service:	Q: Probability of more than	0	customers waiting	= 0%
		Average Utilization of Toilets =	5.22%		
The Total System (waiting line plus customers being served):			Average Number of Customers Being Served (I _p) =	0.8875	
			Average Number in the System (I) =	0.888	
			Average Time in System (T) =	1.5	
Probability Distribution:	n = total number of customers in the system q = number of customers in the waiting line				
n	P(n)	Cumulative	q	P(q)	Cumulative
0	0.4117	0.4117			
1	0.3654	0.7771			
2	0.1621	0.9392			
3	0.0480	0.9871			
4	0.0106	0.9978			
5	0.0019	0.9997			
6	0.0003	1.0000			
7	0.0000	1.0000			
8	0.0000	1.0000			
9	0.0000	1.0000			
10	0.0000	1.0000			
11	0.0000	1.0000			
12	0.0000	1.0000			
13	0.0000	1.0000			
14	0.0000	1.0000			
15	0.0000	1.0000			
16	0.0000	1.0000			
17	0.0000	1.0000	0	1.0000	1.0000

Figure A2 presents the steady-state probability distribution for a finite-capacity queue at Level 2 Male Toilets, computed using an arrival rate of 0.59 and a service rate of 0.67. Each bar indicates the probability of having n customers in the system (both waiting and being served), from zero up to the queue's maximum capacity. Under these conditions, the distribution shows a sharp decline after one or two customers, indicating minimal congestion, while the probability of reaching higher occupancy levels approaches zero very quickly. This reflects efficient service rates relative to arrival patterns, suggesting that significant queuing is uncommon at this facility.

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